

Chapter 137 — Malignant Renal Tumours

Campbell-Walsh-Wein Urology · Chapter Notes

HISTORICAL CONSIDERATIONS

- **First documented nephrectomy — Wolcott, 1861:** performed in error, removing a tumour believed to be a hepatoma.
- **First planned nephrectomy — Simon, 1869:** for a persistent ureteral fistula; the patient survived with cure of the fistula. **First planned nephrectomy in the United States — Gilmore, 1870** (Mobile, Alabama), for atrophic chronic pyelonephritis.
- **Harris, 1882:** reported 100 renal extirpations, the first series large enough to analyse the clinical and pathological features of surgical renal disease.
- **Tubular origin recognised early:** first accurate gross description of kidney tumours by König (1826); Robin (1855) concluded that renal carcinoma arises from **renal tubular epithelium**, supported by Waldeyer (1867).
- **Grawitz (1883) — the adrenal-rest error:** proposed that clear-cell renal tumours arise from **adrenal rests** within the kidney (*struma lipomatodes aberrata renis*), reasoning from their fat content, subcapsular location, proximity to the adrenal and dissimilarity to uriniferous tubules.
- **"Hypernephroma":** the term *hypernephroid tumours* (origin above the kidney) was introduced in 1894; this semantic and conceptual mistake produced the long-lived term **hypernephroma**. Albarran and Imbert (1903) and Wolff's four-volume work (1883–1928) later clarified the histopathology.
- **Modern era:** renal cell carcinoma (RCC) is recognised as **a family of distinct subtypes arising from different parts of the nephron**, each with its own genetic basis and biology.
- **Surgical evolution: classic radical nephrectomy (RN) introduced in the 1960s**, followed by a trend towards **nephron-sparing surgery** and minimally invasive approaches.
- **RCC remains primarily a surgical disease:** immune-based and targeted molecular therapies occasionally give durable responses, but **cure is rarely seen without complete surgical excision or effective ablation**.
- **Incidence is gradually rising**, and despite earlier detection, **mortality remains high**.

Concept — *The history is a correction of an anatomical error: once RCC was understood as tubular, and later as several distinct nephron-derived diseases, management moved from the uniform radical operation towards tailored, nephron-sparing treatment.*

CLASSIFICATION

- Renal masses are classified two ways: **by pathology** (malignant, benign, inflammatory — Table 137.1) or **by radiographic appearance** (simple cystic, complex cystic, solid — Table 137.2). Both schemes have been updated to reflect the distinct RCC subtypes.
- **Malignant renal tumours:** RCC, urothelium-based malignancies, sarcomas, embryonic or paediatric tumours, lymphomas and metastases. **Benign tumours** are diverse (Chapter 136). **Inflammatory and vascular entities** must also enter the differential diagnosis.

TABLE 137.1 RENAL MASSES CLASSIFIED BY PATHOLOGIC FEATURES

Category	Entities
Malignant — renal cell tumours	Clear cell RCC; multilocular cystic renal neoplasm of low malignant potential; papillary RCC; chromophobe RCC; other low-risk oncocytic tumours; RCC not otherwise specified

Category	Entities
Malignant — molecularly defined RCC	TFE3- and TFEB-rearranged RCC; ELOC-mutated RCC; fumarate hydratase-deficient RCC; succinate dehydrogenase-deficient RCC; ALK-rearranged RCC; SMARCB1-deficient renal medullary carcinoma
Malignant — other renal tumours	Collecting duct carcinoma; mucinous tubular and spindle cell carcinoma; tubulocystic RCC; acquired cystic disease-associated RCC; eosinophilic solid and cystic RCC; clear cell papillary renal tumour
Malignant — urothelium-based	Urothelial carcinoma; squamous cell carcinoma; adenocarcinoma
Malignant — sarcomas	Leiomyosarcoma; liposarcoma; Ewing sarcoma; solitary fibrous tumour; others
Malignant — nephroblastic*	Nephrogenic rests; nephroblastoma (Wilms tumour); cystic partially differentiated nephroblastoma; paediatric cystic nephroma
Malignant — paediatric mesenchymal*	Ossifying renal tumour of infancy; congenital mesoblastic nephroma; rhabdoid tumour; clear cell sarcoma of the kidney
Malignant — neuroendocrine	Carcinoid (low-grade); high-grade neuroendocrine carcinoma; neuroblastoma; pheochromocytoma
Malignant — other	Lymphoma, leukaemia, plasmacytoma; germ cell tumours (teratoma, choriocarcinoma); epithelioid angiomyolipoma; metastasis to the kidney; invasion by an adjacent neoplasm
Benign — cystic	Simple cyst; haemorrhagic cyst
Benign — solid	Angiomyolipoma; oncocytoma; papillary adenoma; metanephric tumours; mixed epithelial stromal tumour (adult-type cystic nephroma); reninoma (juxtaglomerular cell tumour); renomedullary interstitial tumour; leiomyoma; haemangioblastoma; schwannoma
Benign — vascular	Renal artery aneurysm; arteriovenous malformation; haemangioma; lymphangioma
Inflammatory / pseudotumour	Abscess; focal pyelonephritis; xanthogranulomatous pyelonephritis; infected cyst; tuberculosis

*More common in childhood or young adults. Modified from Trpkov et al., GUPS update on renal neoplasia, Mod Pathol 2021.

- **What the table is for:** it maps the **2022 World Health Organization (WHO) / Genitourinary Pathology Society framework**, whose main novelty is a separate **molecularly defined RCC group**.
- **Trap:** **epithelioid angiomyolipoma sits among the malignant entities**, whereas classic angiomyolipoma is benign; **papillary adenoma and oncocytoma are benign**.

Imaging and Clinical Risk Stratification of Renal Masses

- Clinical evaluation estimates the **risk of malignancy and metastatic potential** in a given patient: patient characteristics, mode of presentation, imaging, laboratory tests and **renal mass biopsy (RMB)**.
- **Composite models and nomograms predict malignancy poorly** — concordance indices only **0.55–0.65**.
- **Strongest predictors of malignancy: male sex and increasing tumour size**. Males carry a **nearly threefold** risk of malignancy compared with females (meta-analysis of resected tumours). Benign histology is commoner in females, but **RCC still predominates in both sexes**.
- **Size and benignity:** **20–30% of masses ≤4 cm and about 40% of masses ≤2 cm are benign**. The odds of malignancy **rise with each centimetre** of diameter.
- **Size and aggressiveness:** tumours **≤1 cm** are frequently benign and rarely high-grade or high-stage, whereas at **≥7 cm** few are benign and about half are high-grade.
- **Many cT1a (≤4 cm) cancers are indolent:** **<2%** present with or develop metastasis in surgical series, and **≤2% progress to metastasis** on active surveillance (AS) in the short to intermediate term.

- **Complex cystic masses and cystic RCC have an excellent prognosis** and a diminutive metastatic risk, whatever the management strategy chosen.
- Because size drives risk and most masses are found incidentally, **risk stratification begins with imaging**.

Radiographic Evaluation of Renal Masses

- **Imaging is the strongest predictor of malignancy and metastatic potential.** Guidelines recommend **multiphase cross-sectional imaging** for every suspected renal mass, because of the wide differential diagnosis and the overlap between benign and malignant lesions.
- **Intravenous pyelography and renal arteriography are no longer recommended.** Multiphasic cross-sectional imaging characterises the mass, detects locally advanced features and intra-abdominal metastases, and **excludes angiomyolipoma (AML) by showing intralesional fat**.

COMPARISON — IMAGING MODALITIES FOR THE RENAL MASS

Modality	Role	Key threshold or feature	Limitation
Dedicated renal computed tomography (CT) , without and with contrast	Single most important test	Enhancement >15–20 Hounsfield units (HU) → RCC (does not exclude benign); fat below –20 HU → diagnostic of AML	Contrast risk in chronic kidney disease (CKD)
Magnetic resonance imaging (MRI) with gadolinium	Alternative standard; tissue planes and extent of venous involvement	Enhancement >20% suggests RCC; helpful in masses <2 cm	Nephrogenic systemic fibrosis (NSF) with older agents
Ultrasound	Cystic versus solid; inexpensive	Strict simple-cyst criteria (below)	Anything not a simple cyst needs cross-sectional imaging
Contrast-enhanced ultrasound (CEUS) , microbubbles	Enhancement assessment when CT/MRI contraindicated	—	Reasonable alternative, not first choice
Fluorodeoxyglucose positron emission tomography (FDG-PET)/CT	Limited role in RCC	—	High background renal uptake; low sensitivity
Technetium-99m-sestamibi single-photon emission CT (SPECT)/CT	Distinguishes oncocytoma and hybrid oncocytic/chromophobe tumours from clear cell RCC	Strong uptake (mitochondria-rich) → oncocytic tumour	Still under evaluation
Iodine-124-girentuximab PET	Targets carbonic anhydrase IX (clear cell RCC)	REDECT trial defined baseline performance; ZIRCON trial will define utility	In development
Prostate-specific membrane antigen (PSMA) imaging; multiparametric MRI; radiomics and artificial intelligence	Evolving	—	Not yet in routine practice

- **Contrast and renal function:** patients with estimated glomerular filtration rate (eGFR) **<45 mL/min/1.73 m²** receiving intravenous CT contrast should be considered for **periprocedural hydration**; in **severe CKD**, iodinated contrast is avoided where possible.

- **NSF** was commonest on dialysis, in CKD stages 4–5, in acute kidney injury and after multiple doses of **earlier-generation gadolinium agents**. **Newer macrocyclic gadolinium agents carry a very low NSF risk**, allowing contrast MRI even on dialysis or in severe CKD.
- **Strict ultrasound criteria for a simple cyst: smooth wall; round or oval shape without internal echoes; through-transmission with strong posterior acoustic enhancement**. Anything that fails these criteria proceeds to multiphasic cross-sectional imaging.
- **Sestamibi mechanism**: uptake by **mitochondria**, abundant in oncocytomas; combined with selective RMB, it offers a **cost-effective way to avoid surgery for benign tumours**.
- **Every imaging report should state**: diameter in three planes; morphology and relation to hilum, renal vein and collecting system; enhancement; lymphadenopathy; abdominal metastases.
- **Solid architecture predicts malignancy** more than cystic, although many complex cystic masses are RCC.
- **Tumour complexity scores** — **R.E.N.A.L. nephrometry score** (Radius, Endophytic/exophytic, Nearness to collecting system, Anterior/posterior, Location relative to polar lines), **PADUA score** and **C-index** — guide **partial versus radical nephrectomy, open versus minimally invasive approach, and complication risk**; higher complexity may also correlate with malignancy and aggressive histology.
- **Infiltrative growth broadens the differential**: urothelial carcinoma, lymphoma, infection, high-grade or sarcomatoid RCC.
- **Papillary RCC (PRCC) is often hypoenhancing** relative to other subtypes, but enhancement pattern alone **cannot define biology** — clear cell RCC, oncocytoma and PRCC overlap considerably.

Figure 137.1 — Algorithm for Radiographic Evaluation of a Renal Mass

- **Simple cyst (water density)** → observe unless symptomatic.
- **Complex cyst or solid mass, no significant contrast allergy**:
- **eGFR >45** → dedicated renal CT with and without contrast, or MRI with and without gadolinium.
- **eGFR 30–45** → periprocedural hydration, then the same studies.
- **eGFR <30** → **non-contrast CT or MRI with gadolinium** (preferred over ultrasound or CEUS, which remain reasonable alternatives). **Macrocyclic gadolinium is safe in CKD and on dialysis**.
- **Significant contrast allergy** → ultrasound or CEUS, or MRI with or without gadolinium.
- **Interpretation**: simple cyst → observe; complex cyst → define malignant risk; **fat (–20 HU or less) → likely AML**; solid enhancing mass → consider management options and the role of RMB.

TABLE 137.2 RADIOLOGIC AND PATHOLOGICAL CORRELATES FOR RENAL MASSES

Imaging pattern	Principal entities
Simple cystic	Benign cyst; parapelvic cyst; hydronephrosis; calyceal diverticulum
Complex cystic	Cystic RCC; haemorrhagic or hyperdense cyst; benign complex cyst; cystic nephroma; mixed epithelial–stromal tumour; cystic Wilms tumour; infected cyst or abscess; hydrocalyx; arteriovenous malformation; renal artery aneurysm
Strongly enhancing	Clear cell RCC ; AML; oncocytoma ; papillary or chromophobe RCC occasionally
Moderately enhancing solid	Papillary RCC ; chromophobe RCC ; oncocytoma; fat-poor AML; adenoma or metanephric adenoma; unifocal lymphoma; sarcoma; lobar nephronia; pseudotumour; infarct; metastasis
Infiltrative	Lymphoma ; high-grade urothelial carcinoma ; sarcomatoid differentiation ; collecting duct carcinoma ; renal medullary carcinoma ; xanthogranulomatous pyelonephritis; metastasis occasionally
Calcified	RCC; urothelial carcinoma; benign complex cyst; xanthogranulomatous pyelonephritis; renal artery aneurysm; nephrolithiasis

Imaging pattern	Principal entities
Fat-containing	AML; liposarcoma; lipoma
Multifocal / bilateral	Familial RCC; metastases; sporadic multifocal RCC (especially papillary or clear cell); AMLs (tuberous sclerosis); lymphoma; cystic tumours (autosomal dominant polycystic kidney disease)

Modified from Simmons, Herts and Campbell, AUA Update Series, 2007.

- **What the table tests:** the **enhancement gradient** — clear cell and oncocytoma enhance strongly, papillary and chromophobe moderately — and the **infiltrative group, which points away from conventional RCC** and towards lymphoma, urothelial carcinoma or aggressive variants.
- **Trap:** **oncocytoma appears in both enhancing rows** — enhancement cannot separate it from clear cell RCC.

Evaluation of Cystic Renal Lesions

- **Distinguishing benign cysts from cystic RCC is one of the commonest and hardest problems in renal imaging.** Simple cysts carry a near-zero malignant risk; **risk rises with complexity** — wall thickness and contour; number, contour and thickness of septa; amount, character and location of calcification; fluid density; margination; and **solid components**.
- **The Bosniak classification** (CT-based) sorts cystic lesions into categories distinct in malignant likelihood. A **2019 update proposal** added detailed definitions and **incorporated MRI**, aiming to reduce inter-reader variability; its impact is not yet proven, and the fundamental concepts are **unchanged since the 2005 revision**.

TABLE 137.3 CLASSIFICATION OF COMPLEX RENAL CYSTS (BOSNIAK)

Bosniak	Radiographic features	Risk of malignancy	Management
I	Water density; homogeneous, hairline-thin wall; no septa, calcification or enhancement	None	Surveillance not necessary
II	Few hairline septa (perceived enhancement may be present); fine or short-segment slightly thickened calcification; no unequivocal enhancement	Minimal	Surveillance not necessary
II — hyperdense	≤3 cm, well marginated, no unequivocal enhancement	Minimal	Periodic surveillance (table); see note
IIF	Multiple hairline septa; minimal smooth wall thickening; perceived enhancement may be present; thick or nodular calcification without enhancement ; generally well marginated	3–5%	Periodic surveillance
IIF — hyperdense	>3 cm or totally intrarenal, no enhancement	5–10%	Periodic surveillance
III	Indeterminate; thickened irregular or smooth walls or septa with measurable enhancement	About 50%	Surgical excision*
IV	Criteria of III plus enhancing soft-tissue components	75–90%	Surgical excision*

*Excision if in good health; AS is an option for selected patients; thermal ablation is probably best avoided for cystic lesions because of concern about tumour spillage. Modified from Israel and Bosniak, 2005.

- **Body text versus table:** the text gives **Bosniak IIF malignancy as 3–10% overall**, with a **15% risk of radiographic progression** to a more complex category; the table splits IIF into 3–5% (septated) and 5–10% (hyperdense >3 cm or intrarenal). The text states that Bosniak II lesions need no surveillance, whereas the table assigns periodic surveillance to the hyperdense ≤3 cm Bosniak II subgroup — **the difference is the subgroup, not the category**.

- **Bosniak I** — by far the commonest cystic lesion; **no treatment or surveillance** if asymptomatic.
- **Bosniak II** — minimally complex: non-enhancing septated cysts, cysts with mural or septal calcium, infected cysts and **hyperdense cysts** (classically **<3 cm**, pre-contrast attenuation **>20 HU** from old or clotted blood).
- **The key discriminator is unequivocal contrast enhancement — absent in II and IIF.**
- **Bosniak III** — cannot be confidently distinguished from malignancy: thickened irregular walls or septa, or any measurable enhancement. **Bosniak IV** — large cystic components, shaggy margins and, above all, **enhancing solid portions, which make the diagnosis of malignancy.**
- **Surgical exploration is often indicated for healthy patients with Bosniak III or IV** unless trauma or infection explains the findings.
- **RMB or aspiration is rarely used for cystic lesions:** sampling error, poor diagnostic yield and **risk of tumour spillage.**
- **Paradigm shift:** surgery for Bosniak III and IV **is no longer mandatory.** Complex cystic masses are **indolent, grow slowly, rarely progress through the Bosniak categories, and do well whatever the treatment;** cystic renal cancers fare **better than size-matched solid masses**, so **AS and alternatives to surgery suit many patients.**
- **Trap:** cystic masses must be distinguished from **necrotic solid cancers that mimic cysts**, such as high-grade papillary subtypes.

Concept — Risk stratification of a renal mass rests on two imaging variables, **size and enhancement**; everything else — the Bosniak categories, the fat rule for AML, the complexity scores — refines those two, and the recognition that small and cystic masses are mostly indolent is what now licenses surveillance.

INCIDENCE

- **RCC accounts for 2–3% of adult malignancies and is the most lethal of the common urological cancers.**
- About **76,000 new cases and 16,000 deaths yearly in the United States**; about **430,000 cases and 130,000 deaths worldwide.**
- About **16 new cases per 100,000 per year**; **male-to-female ratio 2:1.**
- **Disease of older adults:** typical presentation **55–75 years**; median age at diagnosis worldwide about 75 years. Diagnosis is **rising fastest in those under 40.**
- **Ethnic and geographic variation:** incidence and death rates about **50% lower in Asian-American/Pacific Islander** and **5–10% higher in African American** populations (cause unknown). Worldwide mortality is **highest in Europe** and lowest in South Central Asia, Melanesia and Middle Africa.
- **Sporadic in the great majority; only 4–6% familial.**
- **Incidence has risen about 3% per year since the 1970s**, largely from **incidental detection on ultrasound and CT**, with more localised tumours and better 5-year survival. **Stage migration has plateaued**, with about **70% organ-confined** at diagnosis.
- **Mortality peaked in 2002 (3.8 per 100,000)** and fell to about **3.1 per 100,000 by 2018**, credited to **targeted and immunomodulatory systemic therapy** and to overdiagnosis of small cT1a masses.
- **Paediatric renal tumours** are about **7% of childhood cancers**; about **90% are Wilms tumours**, and RCC is only **1–6%.** RCC equals Wilms tumour in frequency in the second decade. Mean age of paediatric RCC is 8–9 years, with **no sex difference.**
- **RCC in children and young adults** is more often **symptomatic, locally advanced, high grade and of unfavourable subtype.** The commonest subtype carries **TFE3 translocations (Xp11)**, producing TFE3

overexpression and mixed clear and papillary features; **up to 15% follow chemotherapy** for another malignancy.

- **Stage for stage, the young respond well to surgery**, with long-term survivors even after RN and lymphadenectomy for node-positive disease, so **formal lymphadenectomy at RN is recommended when RCC is suspected in a child or young adult**.

Concept — The epidemiology is dominated by **incidental detection**: incidence has risen far faster than mortality because imaging finds small, often indolent tumours — the same fact that underlies the modern emphasis on avoiding overtreatment.

AETIOLOGY

- **Cell of origin**: clear cell and papillary RCC arise from the **proximal convoluted tubule**; **chromophobe RCC (ChRCC)** and **collecting duct carcinoma** arise from the **distal nephron**.

COMPARISON — AETIOLOGICAL RISK FACTORS FOR RCC

Factor	Strength	Key facts
Tobacco	Most generally accepted environmental risk factor ; modest relative risk (about 1.4 current, 1.2 former smokers)	All forms implicated; dose-dependent ; risk falls after cessation (supports causation); accounts for 20–30% of RCC in males, 10–20% in females
Obesity	Major; relative risk 1.07 per unit of body mass index (BMI)	>40% of United States RCC may be linked to obesity; " obesity paradox " — higher BMI but lower stage at presentation
Hypertension	Third major factor	The disease, not its treatment , appears responsible (diuretics once implicated, especially for PRCC)
CKD, end-stage renal disease (ESRD), transplantation	Increased diagnosis and mortality	Causal link not established
Trichloroethylene, perchloroethylene	Twofold to sixfold in some studies	Possibly bias
Occupational (metal, chemical, rubber, printing; asbestos, cadmium)	Slight, not convincing	—
Low socioeconomic status, urban background; Western diet, dairy, coffee or tea	Modest, conflicting	—
Moderate alcohol (≥ 15 g or one drink per day)	May be protective	—
Family history (outside known syndromes)	Twofold to fourfold	—

- **Obesity mechanisms**: increased insulin-like growth factor-1; raised circulating oestrogens and other sex-hormone products; arteriolar nephrosclerosis; local inflammation; oxidative stress; gut microflora producing toxic metabolites.
- **Hypertension mechanisms**: hypertensive renal injury; alterations in the **hypoxia-inducible factor (HIF) pathway**; lipid peroxidation; inflammatory, metabolic or functional tubular changes that increase susceptibility to carcinogens.
- **No specific chemical agent is established as causative in humans**, despite viruses, lead compounds and more than 100 chemicals (including aromatic hydrocarbons) in animal models.

KEY POINTS: INCIDENCE AND AETIOLOGY OF RENAL CELL CARCINOMA

- RCC accounts for **2–3% of adult malignancies**, male-to-female **2:1**.
- RCC typically presents between **55 and 75 years**.
- Incidence is rising, **largely from incidental detection**.
- **Strongest risk factor: tobacco**, relative risk **1.4- to 2.5-fold** versus controls.
- **Hypertension and obesity** are also associated with RCC.
- About **4–6% of RCC is familial**.

Familial Renal Cell Carcinoma and Molecular Genetics

- Since the early 1990s, several **familial RCC syndromes** have been identified, and the tumour suppressor genes and oncogenes driving both sporadic and familial RCC characterised. This drove **revision of the histological classification** and the development of **targeted agents and checkpoint inhibitors**.
- **Knudson two-hit hypothesis** (from retinoblastoma): **both alleles of a tumour suppressor gene must be inactivated**. Familial patients are **born with one mutant allele in every cell**, so tumours are **early-onset and multifocal**; sporadic tumours need **two somatic hits in one cell**, so they are **late-onset and unifocal**. The hypothesis holds for RCC, and familial kindreds enabled **linkage analysis**.

TABLE 137.4 FAMILIAL RENAL CELL CARCINOMA SUBTYPES (REBUILT)

Syndrome	Gene (chromosome)	Renal tumour	Other major manifestations	Renal management	Therapeutic target
Von Hippel-Lindau (VHL)	VHL (3p25)	Clear cell RCC, often multifocal	Retinal angiomas; CNS haemangioblastomas; pheochromocytoma	AS <3 cm; excise ≥3 cm, nephron-sparing	HIF-VEGF pathway
Hereditary papillary renal carcinoma (HPRC)	MET (7q31) — oncogene	Multiple bilateral type 1 PRCC	None usually	AS <3 cm; excise ≥3 cm, nephron-sparing	MET kinase
Hereditary leiomyomatosis and RCC (HLRCC)	FH (1q42–43)	Aggressive, often eosinophilic papillary RCC; collecting duct carcinoma; low-grade variants in children	Cutaneous and uterine leiomyomas; uterine leiomyosarcoma	Prompt excision; PN only if wide margins achievable	HIF-VEGF; antioxidant response; reductive carboxylation
Succinate dehydrogenase-deficient RCC (SDH-RCC)	SDHA, SDHB (1p36.13), SDHC (1q23.3), SDHD (11q23.1), *SDHAF2*	Chromophobe, clear cell, papillary or oncocytoma-like; variable aggressiveness	Paragangliomas; papillary thyroid carcinoma	Excision, PN preferred; wide margins if aggressive-looking	HIF-VEGF; reductive carboxylation

Syndrome	Gene (chromosome)	Renal tumour	Other major manifestations	Renal management	Therapeutic target
Birt-Hogg-Dubé (BHD)	Folliculin <i>FLCN</i> (17p11.2)	Multiple chromophobe, hybrid oncocytic tumours, oncocytomas; occasionally clear cell or papillary	Facial fibrofolliculomas; lung cysts; spontaneous pneumothorax	AS <3 cm; excise ≥3 cm, nephron-sparing	mTOR pathway
PTEN hamartoma tumour syndrome (Cowden)	<i>PTEN</i> (10q23)	Papillary RCC or other	Breast tumours; epithelial thyroid carcinoma	AS <3 cm; excise ≥3 cm, nephron-sparing	mTOR pathway
Tuberous sclerosis complex (TSC)	<i>TSC1</i> (9q34) / <i>TSC2</i> (16p13.3)	Multiple AMLs; clear cell RCC (2–3%); cysts / polycystic disease	Cardiac rhabdomyomas; cutaneous angiofibromas; pulmonary lymphangiomyomatosis; autism and cognitive disability	AML: surveillance <3 cm; everolimus 3–5 cm; embolisation or excision ≥5 cm. RCC: excise if ≥3 cm	mTOR pathway
BAP1 tumour predisposition	<i>BAP1</i> (3p21.2)	Clear cell RCC, can be high grade	Uveal melanoma, cutaneous melanoma, mesothelioma	Excision, nephron-sparing	To be determined
MitF-associated cancer syndrome	<i>MitF</i> (3p14.1–p12.3)	Not defined	Melanoma	To be determined	To be determined

mTOR, mammalian target of rapamycin; VEGF, vascular endothelial growth factor. Modified from Schmidt and Linehan, *Semin Oncol* 2016.

- The pattern: 3 cm is the intervention threshold for the indolent syndromes (VHL, HPRC, BHD, Cowden); HLRCC is the exception — excise promptly at any size, with wide margins.
- All familial RCC syndromes are autosomal dominant. Most are tumour suppressor losses; HPRC (*MET*) is the one proto-oncogene activation.

Von Hippel-Lindau Disease, *VHL* Gene and Genetics of Clear Cell RCC

- VHL disease is the familial form of clear cell RCC (CCRCC): autosomal dominant, about 1 in 36,000.
- Major manifestations: RCC, pheochromocytoma, retinal angiomas, and haemangioblastomas of the brainstem, cerebellum or spinal cord — all highly vascular. CNS lesions can cause paralysis or death; retinal lesions, blindness. Others: renal and pancreatic cysts, inner-ear (endolymphatic sac) tumours, pancreatic neuroendocrine tumours, epididymal papillary cystadenomas.
- Penetrance high (70–87%) but incomplete for individual traits; pheochromocytoma clusters in particular families.
- RCC develops in about 50%, with early onset (third to fifth decades), bilateral and multifocal. With better CNS management, RCC is now the commonest cause of death in VHL.

- **Gene discovery:** cytogenetics showed **loss of chromosome 3** (especially in clear cell RCC); linkage in VHL kindreds identified **VHL (1993)** — **three exons, 213-amino-acid protein**, a tumour suppressor for both sporadic and familial CCRCC.
- **Genotype–phenotype:** missense (type 2) mutations → full-length but non-functional protein → **phaeochromocytoma families; deletions or truncations (type 1) → no phaeochromocytoma.**

TABLE 137.5 MANIFESTATIONS OF VON HIPPEL-LINDAU DISEASE

Tumour	Mean age of onset (range), years	Incidence (%)
Retinal angioma	25 (1–68)	25–60
Endolymphatic sac tumour	22 (12–50)	10–15
Craniospinal haemangioblastoma (overall)	30 (9–70)	60–80
— Cerebellum	33 (9–78)	44–72
— Brainstem	32 (12–46)	10–25
— Spinal cord	33 (11–66)	13–50
Renal cell carcinoma	39 (13–70)	25–75
Renal cysts	39 (13–70)	20–60
Phaeochromocytoma	27 (5–58)	10–25
Pancreatic neuroendocrine tumours	36 (5–70)	11–17
Pancreatic cysts	36 (5–70)	45–75
Epididymal cystadenoma	Unknown	25–60
Broad ligament cystadenoma	Unknown (16–46)	Unknown

Modified from Nielsen et al., *J Clin Oncol* 2016.

- **What the table shows:** **craniospinal haemangioblastoma is the commonest lesion; retinal angioma and phaeochromocytoma appear earliest (20s), RCC latest (about 39 years)** — which is why screening starts in childhood but renal imaging only in adolescence.

TABLE 137.6 MAJOR TUMOUR TYPES OF VHL DISEASE BY SUBTYPE AND MUTATION

VHL subtype	Mutation type	High risk	Low risk
Type 1	Deletions, insertions, truncations, missense	RCC, CNS/retinal haemangioblastoma	Phaeochromocytoma
Type 1B	Contiguous gene deletions encompassing VHL	CNS/retinal haemangioblastoma	Phaeochromocytoma, RCC
Type 2A	Missense (e.g., p.Y98H)	Phaeochromocytoma, CNS/retinal haemangioblastoma	RCC
Type 2B	Missense (e.g., p.R167Q)	RCC, phaeochromocytoma and haemangioblastoma — all	—
Type 2C	Missense (e.g., p.V84L)	Phaeochromocytoma only	RCC absent; haemangioblastoma

A missense mutation is a single-nucleotide change altering one amino acid (Y98H: tyrosine at position 98 replaced by histidine).

Modified from Nielsen et al., 2016.

- **The rule:** **type 1 = no phaeochromocytoma; type 2 = phaeochromocytoma.** Within type 2, **2A low RCC, 2B high RCC, 2C phaeochromocytoma alone.**
- **VHL protein function:** binds **elongins B and C** (with cullin-2 and RBX1) to form an **E3 ubiquitin ligase** that targets **HIF-1α and HIF-2α for ubiquitin-mediated degradation**, keeping HIF low in normoxia.

- **Loss of VHL → accumulation of HIF, chiefly HIF-2α → several-fold upregulation of VEGF** (the primary angiogenic factor in RCC), explaining the **neovascularity of CCRCC**. HIF-2α also upregulates **transforming growth factor-α (TGF-α)**, **platelet-derived growth factor (PDGF)**, **glucose transporter 1**, **erythropoietin** and **carbonic anhydrase IX**.
- **Belzutifan (HIF-2α inhibitor)** is approved by the **United States Food and Drug Administration (FDA)** for **VHL syndrome**, inhibiting tumour growth in the great majority.
- **Chromosome 3p is altered in >90% of CCRCC**. Besides *VHL*, the other three commonly mutated 3p genes — *PBRM1*, *BAP1*, *SETD2* — are **chromatin-remodelling / histone-methylation genes**.
- **BAP1 mutations occur in up to 14% of sporadic CCRCC** and carry **poor prognosis**; germline *BAP1* should be considered in **early-onset multifocal CCRCC that is VHL wild-type***. *BAP1-mutant tumours grow rapidly — treat aggressively; not suitable for AS.**

Figure 137.7 — The VHL Pathway and Its Drug Targets

Target	Mechanism	Agents
VEGF ligand	Sequestration	Bevacizumab
VEGF and PDGF receptors	Tyrosine kinase inhibition (TKI)	Sunitinib, sorafenib, pazopanib, axitinib, cabozantinib, lenvatinib (tivozanib also named in text)
mTOR	Reduces translation of new HIF protein	Temsirolimus, everolimus
HIF-2α	Blocks HIF-2α-ARNT dimerisation	Belzutifan (PT2977); earlier PT2385, PT2399

- **Normoxia versus hypoxia or VHL loss:** in normoxia, VHL marks HIF for destruction; in hypoxia or with *VHL* loss, HIF is **stabilised, dimerises with ARNT**, and drives VEGF (angiogenesis), PDGF and TGF-α (glucose transport, autocrine and paracrine growth).

Hereditary Papillary Renal Carcinoma

- **Non-clear-cell RCC rarely shows chromosome 3 or VHL abnormalities**, implying a separate genetic basis. **PRCC — the second commonest subtype — shows trisomy 7 and 17**, plus changes in chromosomes 1, 12, 16, 20 and Y.
- **HPRC (described 1995): median age about 45, multifocal bilateral PRCC**; reported in only about 35 families worldwide (incidence <1 in 1,500,000). **No tumours in other organs**, unlike VHL.
- **Type 1 versus type 2 PRCC** was formerly used (type 1: sporadic/HPRC; type 2: HLRCC) but **has been abandoned** because type 2 was heterogeneous.
- HPRC tumours are type 1-like, **less aggressive → favourable course; avoid overtreatment**. CKD from parenchymal replacement and repeated interventions contributes to morbidity.
- **Mechanism: activating missense mutations of the c-MET proto-oncogene (7q31) — the receptor tyrosine kinase for hepatocyte growth factor — mostly in the tyrosine kinase domain**, causing **constitutive activation**. Inheritance of the mutation puts every renal cell at risk from birth (early onset, multifocality); incomplete penetrance implies modifying loci.
- *c-MET* mutations are also found in a subset of **sporadic PRCC**; **MET inhibitors are in early trials**.

Hereditary Leiomyomatosis and Renal Cell Carcinoma

- **Cutaneous and uterine leiomyomas with aggressive PRCC-like tumours**; mean age at diagnosis early 40s. Renal tumours are **often solitary and unilateral** (unusual for familial RCC) and **more aggressive** than other familial RCC; **collecting duct carcinoma** also occurs.

- **Gene: fumarate hydratase (FH, 1q42–43)** — a Krebs-cycle enzyme, acting as a **tumour suppressor**; germline mutations in **90%** of HLRCC; biallelic somatic loss occurs in sporadic cases. Mechanism relates to the **Warburg effect**.
- **Nomenclature:** "HLRCC syndrome-associated RCC" when familial; "**FH-deficient RCC**" when histological or clinical diagnosis is uncertain.
- **Penetrance for RCC only 15–20%**, whereas **almost all develop cutaneous leiomyomas and (if female) uterine fibroids**, usually at 20–35 years; many women have had a **hysterectomy before the syndrome is recognised**; leiomyosarcoma occasionally.
- **Prompt surgical management of renal tumours is recommended** — in contrast to the conservative approach in most other syndromes. **FDG-PET** can detect small metastatic foci (high metabolic activity). Some young FH carriers have **low-grade oncocytic tumours** resembling SDH-RCC.

Succinate Dehydrogenase-Deficient RCC

- **Bilateral, multifocal, early-onset (≤40 years)** renal tumours, often with **phaeochromocytoma, head and neck paraganglioma and gastrointestinal stromal tumour**.
- **Germline mutations in Krebs-cycle SDH subunit genes (SDHA, SDHB, SDHC, SDHD, SDHAF2)** — another Warburg-effect cancer.
- **Early reports suggested about one-third metastasise**; more recent series show a **good prognosis when low-grade**. High grade, coagulative necrosis or sarcomatoid features predict aggression.

Tuberous Sclerosis Complex

- **Autosomal dominant**; AMLs, cortical tubers, subependymal nodules or subependymal giant cell astrocytomas, pulmonary lymphangioleiomyomatosis, cardiac rhabdomyomas and facial angiofibromas.
- **Renal AMLs are multifocal, bilateral and often large**; angioembolisation or partial nephrectomy (PN) may be needed to treat or prevent bleeding from tumours **≥5 cm**.
- **Genes: TSC1 (9q34, hamartin) or TSC2 (16p13, tuberin)**. The TSC1/TSC2 complex **downregulates mTOR**; mTOR inhibitors (everolimus) are active against TSC-associated AMLs.
- **RCC, typically clear cell, in 1–3%**.

Birt-Hogg-Dubé Syndrome

- **Cutaneous fibrofolliculomas, lung cysts, spontaneous pneumothoraces** and renal tumours of **distal-nephron origin**; named after three **Canadian physicians (1977)**.
- **Tumours: ChrRCC, oncocytomas and hybrid oncocytic tumours**; occasionally CCRCC.
- **Renal penetrance 23–34%**, but tumours are **often bilateral and multifocal**; prevalence about 2 per million; **mean age at renal diagnosis about 50**. Mostly **limited aggressiveness**, though metastasis and death are reported.
- **Gene: folliculin (FLCN, 17p11.2)**, a tumour suppressor interfacing with **mTOR**; germline mutation in **88% of kindreds**. Loss of folliculin activates **mTORC1 and mTORC2**, increasing **TFE3 nuclear translocation**. Autosomal dominant; genetic testing available.

Cowden Syndrome

- One of the **PTEN hamartoma tumour syndromes** (germline **PTEN** mutation).
- **Lifetime risks: female breast cancer 50%, RCC 34%, epithelial thyroid carcinoma 10%**.
- Patients with Cowden features but no **PTEN** mutation may carry **KILLIN** mutations (adjacent tumour suppressor), also raising RCC risk.

- **>31-fold increased RCC risk → screen with CT or MRI.** RCC is **most often papillary**; chromophobe and clear cell also reported.

BAP1 RCC Susceptibility

- **Autosomal dominant:** RCC, melanocytic tumours, **cutaneous or uveal melanoma, malignant mesothelioma.**
- **BAP1 is a nuclear deubiquitinating enzyme** with chromatin-modifying function like PBRM1 and SETD2 — all on chromosome 3.
- **BAP1 and PBRM1 have distinct signatures; BAP1 mutation associates with necrosis, higher grade and worse survival.**

MiTF-Associated Cancer Syndrome

- A germline **missense variant (p.E318K) of microphthalmia-associated transcription factor (MiTF)** confers a **>5-fold risk of melanoma and RCC.**
- MiTF regulates **HIF-1 α** ; the mutation activates HIF-1 α , potentially promoting RCC.

Concept — Every familial syndrome converges on a small number of pathways — **HIF/VEGF (VHL), MET, Krebs-cycle metabolism (FH, SDH) and mTOR (TSC, BHD, PTEN)** — and the pathway predicts both the histology and the drug; the one clinical outlier is HLRCC, where the tumour's aggression overrides the 3 cm surveillance rule.

Tumour Biology and Clinical Implications

Resistance to Cytotoxic Therapy

- **Cytotoxic chemotherapy response only 5–7%** — the historical reason for poor metastatic outcomes.
- **Multidrug resistance (MDR) proteins** pump out hydrophobic drugs; RCC also resists **cisplatin, MDR inhibitors and non-MDR substrates**, implying **redundant resistance mechanisms.**
- Understanding the **VEGF and mTOR pathways and checkpoint inhibition** produced effective drugs (Chapter 144).

TABLE 137.7 TUMOUR BIOLOGY AND CLINICAL IMPLICATIONS

Biological characteristic	Clinical implication
Multidrug resistance expression	Chemorefractory
Immunogenic	Historical 10–20% response to interferon or interleukin-2 (IL-2) , now abandoned for toxicity. Checkpoint inhibition (programmed death-1 [PD-1], PD-ligand 1 [PD-L1], cytotoxic T-lymphocyte antigen-4 [CTLA-4]: nivolumab, pembrolizumab, atezolizumab, ipilimumab, avelumab) → strong responses, prolonged recurrence-free and overall survival
Angiogenic	Vascular invasion → venous tumour thrombus . 20–40% responses with VEGF-pathway agents (sunitinib, sorafenib, pazopanib, axitinib, bevacizumab). HIF-2α inhibitor in VHL: 50% response in RCC. Some antiangiogenic agents prolong recurrence-free and overall survival
mTOR dependence	Temsirolimus prolongs survival in poor-risk RCC; everolimus active after failure of prior targeted therapy

Immunobiology and Immune Tolerance

- **Clues to immunogenicity:** tumour-infiltrating immune cells; occasional **spontaneous regression**; **durable complete regression with IL-2 in 3–5%.**

- **Cytokine era:** overall responses only **15–20%**, with **2–3% treatment-related mortality**. Tumour-infiltrating lymphocyte reinfusion and allogeneic stem cell transplantation largely failed.
- **Immune checkpoint inhibitors (ICIs) are now the mainstay of immunomodulation** in advanced RCC.

Immune Checkpoint Inhibition

- **RCC evades immune surveillance** through mechanisms at nearly every level; **T-cell downregulation is central**.
- **CTLA-4 and PD-1** are T-cell glycoprotein receptors that normally **prevent autoimmunity**. **Tumour PD-L1 expression** engages PD-1 → effector T-cell suppression and tolerance. **PD-L1 expression in CCRCC predicts aggressive pathology and progression** independently.

COMPARISON — CHECKPOINT INHIBITORS (FIGURE 137.8)

Target	Interaction blocked	Agents
CTLA-4	B7 on antigen-presenting cell with CTLA-4 on T cell (inhibitory; competes with B7-CD28 activation)	Ipilimumab
PD-1 (on T cell)	Tumour PD-L1 with T-cell PD-1	Nivolumab, pembrolizumab
PD-L1 (on tumour cell)	Same interaction, ligand side	Atezolizumab, avelumab

- **T-cell activation depends on the balance** of positive signals (T-cell receptor with major histocompatibility complex/antigen; B7-CD28) and negative signals (B7-CTLA-4; PD-L1-PD-1). **Blocking the negative side tilts towards activation**.
- **ICIs, combined with TKIs, other ICIs and sometimes surgery, have pushed median survival in metastatic RCC towards 5 years**.
- **Immune-related adverse events: enterocolitis, hepatitis, dermatitis, pneumonitis**.

Angiogenesis and Targeted Pathways

- **RCC is one of the most vascular cancers**. **VEGF**, suppressed by wild-type VHL, is the primary angiogenic inducer in CCRCC; **serum and urine VEGF are higher with hypervascular tumours**.
- **VEGF is a ligand family** regulated by HIF, mTOR and VHL, binding VEGF receptors (VEGFR) → tyrosine phosphorylation → endothelial survival and proliferation. **Promiscuous ligand-receptor interactions make each TKI's effects hard to predict**.
- **Bevacizumab was the first successful agent to target the VHL pathway**.
- **Cabozantinib inhibits VEGF, MET, RET and AXL** — active even second-line, suggesting **multiple pathways must be targeted in refractory disease**.
- **Belzutifan: 50% response in VHL** (renal tumours, retinal haemangioblastomas, pancreatic lesions); **FDA approved 2021 for VHL**; now studied with TKIs and ICIs in sporadic RCC.

Cancer Genome Research

- **The Cancer Genome Atlas (TCGA)** characterised CCRCC in **2013**, and in **2018** published data on **843 RCCs** (488 clear cell, 274 papillary, 81 chromophobe), showing shared and subtype-specific features.
- **16 significantly mutated genes; VHL is the commonest somatic mutation (30–40%)**. *FH* may be more often mutated in the general population but with **low penetrance**. Pathways: chromatin remodelling, VHL/HIF, HIPPO, NRF2/ARE, PI3K/AKT.

COMPARISON — COPY-NUMBER SIGNATURES BY SUBTYPE

Subtype	Signature change
Clear cell	Loss of 3p (VHL locus); gain of 5q
Papillary	Gains of 7 and 17
Chromophobe	Losses of 1, 2, 6, 10, 13, 17

*Molecular Changes Associated With Survival***COMPARISON — GENOMIC MARKERS OF REDUCED SURVIVAL**

Scope	Alteration	Subtypes affected
All major subtypes	CDKN2A alteration, mutation or deletion (9p; encodes cell-cycle regulators p14, p16)	Clear cell, papillary, chromophobe
All major subtypes	DNA hypermethylation — CpG island methylator phenotype (CIMP-RCC); uncommon, extremely poor prognosis	All
All major subtypes	Increased immune-related Th2 gene signature	All
Subtype-specific	BAP1	Clear cell only
Subtype-specific	TP53	Clear cell and papillary, not chromophobe
Subtype-specific	PBRM1	Papillary and chromophobe
Subtype-specific	PTEN	Chromophobe only

- **Metabolic shift in aggressive RCC: downregulated tricarboxylic acid (TCA) cycle genes**, upregulated **pentose phosphate and glutamine transporter** genes, reduced AMPK and PTEN, increased acetyl-CoA carboxylase — **metabolic remodelling as a future therapeutic window**.

Other Signal Transduction and Cell Cycle Regulation Pathways

- **mTOR** interfaces with **Akt** and **PTEN**; upregulated by growth factors or **PTEN loss**, and **increases HIF expression**. **Temsirolimus and everolimus work after TKI failure**, confirming relevance.
- **Hepatocyte growth factor / c-MET**: normally drives branching tubulogenesis and tubular regeneration; **upregulated across most RCC subtypes** (not only hereditary PRCC); serum hepatocyte growth factor is raised in most RCC; receptor phosphorylation correlates with progression — **c-MET is a target for papillary and other RCC**.

Microbiome

- **Microbes outnumber human cells by nearly 2:1**.
- Evidence in RCC: certain microbiota are **protumorigenic** in immune-responsive tumours; **antibiotics during ICI therapy nearly halved median overall survival** in one study; **responders and non-responders to ICI have different microbial fingerprints**; bacterial strains alter anti-CTLA-4/anti-PD-1 responses in mice; **targeted antibiotics against deleterious species improved response**; tumour and normal renal tissue carry different microbiota.

Concept — RCC's biology explains its therapeutic history: **multidrug resistance** made chemotherapy fail, **VHL-driven hypervascularity** made anti-angiogenic drugs work, and **immune evasion through checkpoints** made checkpoint blockade the modern mainstay.

Pathology

- **Most RCCs are round to ovoid with a pseudocapsule** (compressed parenchyma and fibrous tissue), **not a true capsule**. Unlike upper-tract urothelial carcinoma, most are **not grossly infiltrative** — exceptions: **collecting duct carcinoma and sarcomatoid variants**.
- **Size**: usually 4–8 cm. **Tumours <3 cm are typically indolent RCC or benign**, though small tumours rarely metastasise.
- **Gross**: yellow, tan or brown, with fibrosis, necrosis or haemorrhage. **Cystic degeneration in 10–25%**, associated with **better prognosis**; **calcification in about 10%** (stippled or plaque-like).
- **Nuclear grade is an independent prognostic factor**, based on nuclear size, shape and nucleoli. The **four-tier WHO / International Society of Urological Pathology (ISUP) system** is validated for clear cell and papillary RCC and is the recommended system.

TABLE 137.8 WHO/ISUP GRADING SYSTEM FOR CLEAR CELL AND PAPILLARY RCC

Grade	Description
1	Nucleoli absent or inconspicuous and basophilic at ×400
2	Nucleoli conspicuous and eosinophilic at ×400, visible but not prominent at ×100
3	Nucleoli conspicuous and eosinophilic at ×100
4	Extreme nuclear pleomorphism, multinucleated giant cells, and/or rhabdoid and/or sarcomatoid differentiation

Modified from Delahunt et al., *Eur Urol* 2014.

- **The logic**: grades 1–3 turn on **the magnification at which nucleoli become visible**; grade 4 is **morphological** — **rhabdoid or sarcomatoid differentiation is automatically grade 4**.
- **Aggressive local behaviour in about 20%**: invasion of perinephric fat, renal sinus or collecting system. **Gerota's fascia usually prevents spread** to adjacent organs, although high-grade tumours can breach it.
- **Venous involvement in about 10% — more than any other tumour type**: thrombus into renal vein and inferior vena cava (IVC), **as high as the right atrium**. Thrombi are often **arterially vascularised**; **direct invasion of the venous wall worsens prognosis**.
- **Most sporadic RCCs are solitary**. **Bilateral RCC (synchronous or asynchronous) in 2–4% of sporadic cases**, commoner in familial forms. **Multicentricity in 10–20%**, commoner in **PRCC and familial RCC**.
- **Satellite lesions** are small and may be invisible on imaging, intraoperative ultrasound or inspection — **the main cause of ipsilateral recurrence after PN**.
- **Clonality**: multifocal CCRCC in one kidney is usually **clonal**; a contralateral tumour is likely **independent if synchronous, metastatic if asynchronous**.
- **Intratumour heterogeneity**: multiregion sequencing shows single biopsies may **miss heterogeneity** and confuse driver with passenger mutations — a challenge for biomarkers and personalised medicine.
- **All RCCs are adenocarcinomas** of renal tubular epithelium. Most share proximal-tubule ultrastructure (microvilli, complex junctions); **ChRCC, SMARCB1-deficient medullary carcinoma and collecting duct carcinoma arise distally**.
- **Classification history**: traditionally four subtypes (**clear cell, papillary, chromophobe, collecting duct/medullary**) by cytoplasm and location; **revised by WHO in 2022** to add newly described entities.

TABLE 137.9 PATHOLOGICAL SUBTYPES OF RCC — COMMON SUBTYPES (REBUILT)

Subtype (share)	Genetics / familial form	Gross	Microscopy and immunohistochemistry (IHC)	Origin and behaviour
Clear cell (70–80%)	VHL disease; VHL (3p25) mutation or hypermethylation; 3p loss; loss 8p, 9p, 14q; gain 5q	Golden-yellow, hypervascular, well circumscribed, lobulated; necrosis, haemorrhage, cystic change; venous involvement common	Nests or sheets of clear cells, delicate vascular network ; low-molecular-weight cytokeratins, vimentin, epithelial membrane antigen, carbonic anhydrase IX (CA-IX)	Proximal tubule; more aggressive; responds to TKI and ICI
Papillary (5–10% in table; 10–15% in text)	HPRC; trisomy 7 and 17; MET altered in 81% of sporadic	Fleshy, fibrous pseudocapsule; necrosis and haemorrhage	Hypovascular ; papillae with single cell layer on fibrovascular cores; basophilic low-grade cells; CK7, alpha-methylacyl-CoA racemase (AMACR)	Proximal tubule; good prognosis; often multicentric; common in acquired renal cystic disease; no longer split into types 1 and 2
Chromophobe (3–5%)	BHD; FLCN (17p11)*; loss of chromosomes 1, 2, 6, 10, 13, 17, 21, Y; TP53, PTEN***	Well circumscribed, homogeneous, tan or light brown	"Plant cells": pale cytoplasm, perinuclear halo, nuclear raisins , prominent cell borders; diffuse CK7 ; eosinophilic subtype (mitochondrial gene mutations)	Intercalated cells of collecting duct; good prognosis except sarcomatoid
Clear cell papillary tumour (<5%)	VHL	Well circumscribed, well-developed capsule	Low-grade clear cells in linear papillae; diffuse CK7; cup-like CA-IX	Arises in ESRD and VHL ; now generally considered benign
Unclassified RCC (1–3%)	Unknown; retain SMARCB1	Varied	Varied	Generally poor prognosis

TABLE 137.9 (CONTINUED) — RARE AND MOLECULARLY DEFINED SUBTYPES (REBUILT)

Subtype	Genetics	Distinguishing features	Behaviour
SMARCB1-deficient renal medullary carcinoma	Sickle cell trait	Infiltrative, grey-white, haemorrhage and necrosis; lace-like poorly differentiated cells; IHC: SMARCB1 loss	Collecting duct origin; dismal prognosis
FH-deficient RCC (formerly HLRCC-associated)	FH (1q42–43)	Eosinophilic high-grade papillary cells, macronucleolus with perinucleolar clearing ; IHC: FH loss, strong diffuse 2SC	Metastasises early; extremely poor prognosis; low-grade variant in children
SDH-deficient RCC	SDH genes	Solitary; vacuolated eosinophilic or clear cells, intracytoplasmic inclusions ; IHC: SDHB loss	Young adults, often germline ; aggressive or indolent
Eosinophilic solid and cystic RCC	Copy-number gains 7p–7q, losses Xp11, 16p, Xq	Solid with macro- and microcysts; granular stippled cytoplasm; CK20 positive	Females only; indolent

Subtype	Genetics	Distinguishing features	Behaviour
ALK-rearranged RCC	ALK fusions (2p23)	Eosinophilic, vacuolated, broad morphology; ALK IHC	Heterogeneous; dramatic responses to ALK inhibitors in some
ELOC-mutated RCC (RCC with fibromyxomatous stroma)	Somatic ELOC (formerly TCEB1), TSC1, TSC2, *MTOR*	Fibromuscular septa; diffuse CK7 ; must be distinguished from clear cell papillary tumour	Indolent
MiT family translocation RCC (Xp11/TFE3, TFEB)	Xp11.2 fusions	Tan-yellow; clear cells with papillary architecture ; FISH diagnostic	40% of paediatric RCC ; t(X;17) advanced but often indolent; t(X;1) late nodal recurrence
Acquired cystic disease-associated RCC	VHL alterations, 3p deletion	Cribriform or sieve-like, cysts lined by clear cells	ESRD with acquired cystic disease; excellent prognosis
Collecting duct carcinoma (Bellini)	Multiple chromosomal losses	Firm, central, infiltrative ; desmoplastic stroma, high-grade; PAX8 confirmation	Collecting duct; poor prognosis ; may respond to cytotoxic chemotherapy
Multilocular cystic clear cell neoplasm of low malignant potential	As clear cell	Cysts lined by single layer of low-grade clear cells, no expansile nodules	Almost uniformly benign
Tubulocystic RCC	Unknown	Spongy multicystic; grade 3 nucleoli, oncocytoma-like cells	Favourable
Mucinous tubular and spindle cell carcinoma	Unknown	Medullary ; tubules and spindle cells in mucin	Favourable
Hybrid oncocytic chromophobe tumour	BHD; FLCN	Features of chromophobe and oncocytoma	Good

Sarcomatoid variants of all subtypes occur and worsen prognosis. Molecularly defined RCCs are usually recognised by morphology and confirmed molecularly when necessary. Modified from Moch et al., 2022 WHO classification, Eur Urol 2022.

- **What the table is for:** sorting subtypes into **good-prognosis (papillary, chromophobe, clear cell papillary, multilocular cystic, tubulocystic, mucinous tubular and spindle cell, eosinophilic solid and cystic, ELOC)** and **lethal (medullary, FH-deficient, collecting duct, unclassified, sarcomatoid)**.
- **Discrepancy:** the PRCC share is **5–10% in Table 137.9** but **10–15% in the text and Key Points**; the text figure is the one conventionally taught.

Clear Cell Renal Cell Carcinoma

- **70–80% of RCC — the "garden variety".** Yellow and highly vascular, with delicate sinusoids between sheets or acini of cells.
- **Clear cytoplasm** results from **glycogen, cholesterol, cholesterol esters and phospholipids dissolved** during processing; granular eosinophilic (mitochondria-rich) cells may predominate.
- **Sarcomatoid features in 5–8%; venous extension more likely than any other subtype.**
- **The prognostic paradox:** localised CCRCC does **worse than PRCC or ChRCC** even after stage and grade adjustment, yet **metastatic CCRCC does better**, because it responds to VEGF-targeted therapy and ICIs.
- **3p alterations in >90% (VHL, PBRM1, SETD2, BAP1).**

Papillary Renal Cell Carcinoma

- **Second commonest (10–15%).** Basophilic or eosinophilic cells in papillary or tubular arrangement; gross beige-white, spherical, **frequent haemorrhage that may mimic cystic change** on imaging.
- **Multicentricity approaching 40%** in some series, more so in **ESRD and acquired renal cystic disease**.

- **Type 2 removed from the 2022 WHO classification** as heterogeneous, giving a more homogeneous PRCC category.
- **TCGA: MET altered in 81% (mutation, splice variant, fusion, copy number); trisomy 7 and 17, loss of Y*; also gains of 12, 16, 20 and loss of heterozygosity on 14.** MET is being targeted with cabozantinib.*
- **Hypovascular** relative to clear cell, probably from absent VHL mutation.
- **5-year cancer-specific survival 86–92%**, likely better now aggressive variants are excluded.
- **Papillary adenomas — ≤15 mm, low-grade**, common at autopsy, share PRCC genetics, **considered indolent**.

Chromophobe Renal Cell Carcinoma

- **3–5% of RCC**, from the **distal convoluted tubule**. **Low mutational burden** — only **TP53** and ***PTEN frequent**; **FLCN*** in BHD, but most are sporadic.
- **Classic "plant cell"**: transparent reticular cytoplasm and **prominent cell membrane**; **eosinophilic subtype** about 30%. A **perinuclear halo** suffices for **diagnosis on haematoxylin and eosin**.
- **Hypodiploid**; **whole-chromosome losses (1, 2, 6, 10, 13, 17) in 86%**; **mitochondrial dysfunction** is a common feature.
- **Better prognosis than CCRCC**; **>90% cancer-free at 5 years**; poor outcome with **sarcomatoid features or metastasis**. Aggressive tumours may carry **mTOR-activating mutations**; **TKIs, mTOR inhibitors and ICIs** have **limited activity** in metastatic disease.

Other Oncocytic Renal Neoplasms

- **Chromophobe versus oncocytoma remains hard**, with poor agreement even among expert pathologists.
- **2022 WHO descriptive term: "oncocytic renal neoplasm of low malignant potential, not otherwise specified"** — for **solitary, sporadic** tumours; a **clinical management category**, heterogeneous but **indolent** with **exceedingly low metastatic potential**.

Collecting Duct Carcinoma

- **Carcinoma of the collecting ducts of Bellini** — **rare, poor prognosis**. Mostly large infiltrative masses from a medullary pyramid invading cortex; dilated tubules and papillae lined by cuboidal cells (**cobblestone appearance**).
- **A diagnosis of exclusion**: many historical cases were **FH-deficient or medullary RCC, urothelial carcinoma with glandular differentiation, or metastasis** (e.g., lung). Strict criteria require **PAX8 expression** and exclusion of these — now **vanishingly rare**.
- **High grade, advanced, unresponsive to conventional therapy**; shares features with urothelial carcinoma, so some respond to **cisplatin- or gemcitabine-based chemotherapy**; **multimodality therapy** outperforms nephrectomy alone in metastatic disease.

Molecularly Defined Renal Cell Carcinoma

- **2022 WHO update** adds subtypes redefined **molecularly**, communicating biology more accurately, especially when morphology is ambiguous.
- **Former type 2 PRCC comprised at least three entities: MET-altered tumours; MiT family translocation RCC; FH-deficient RCC.**
- **MiT family (TFE3, TFEB)**: heterogeneous clear-to-eosinophilic, papillary and nested; **commonest RCC in children and young adults**, but similar absolute numbers in older adults; **confirm with TFE3/TFEB FISH**.
- **ALK-rearranged RCC**: very rare, heterogeneous, often called unclassified; **test ALK IHC/FISH before labelling "unclassified"**, especially with myxoid patterns.

- **SDH-deficient RCC:** young adults, usually favourable, but **recognition matters because many are germline** (hereditary paraganglioma spectrum); **SDHB IHC** confirms.
- **ELOC-mutated RCC:** part of "RCC with fibromyxomatous stroma" (with sporadic *TSC1*, *TSC2*, *MTOR*); identified by TCGA; **indolent**.
- **The two aggressive molecularly defined tumours: SMARCB1-deficient medullary RCC and FH-deficient RCC;** both identified by morphology and IHC, with molecular analysis for confirmation.

COMPARISON — MEDULLARY CARCINOMA VERSUS COLLECTING DUCT CARCINOMA

Feature	SMARCB1-deficient medullary RCC	Collecting duct carcinoma
Setting	Sickle cell trait; Black patients, third decade	Older adults; diagnosis of exclusion
SMARCB1	Lost	Retained
Chemotherapy	Platinum: modest objective response; no response to targeted therapy	More likely to respond to cisplatin-based chemotherapy
Prognosis	Median survival about 8–13 months; <15% survive 2 years	Poor

- **SMARCB1-deficient "medullary-like" RCC** also occurs **without sickle trait**, with the same grave prognosis.
- **FH-deficient RCC:** very aggressive, **manage promptly; high-grade eosinophilic, macronucleolus with perinucleolar clearing; FH and 2SC IHC;** mostly HLRCC-associated, some somatic. *FH* mutations may be common in the population but **penetrance is low**.

Sarcomatoid and Rhabdoid Differentiation

COMPARISON — SARCOMATOID VERSUS RHABDOID DIFFERENTIATION

Feature	Sarcomatoid	Rhabdoid
Frequency	1–5% of RCC, mostly clear cell or chromophobe	Seen in many subtypes
Nature	Dedifferentiated region of the parent RCC, not a separate tumour	Not a separate subtype
Histology	Spindle cells , infiltrative growth	Rhabdoid cells
Grade	Grade 4	Mandates grade 4
Prognosis	Median survival <1 year in most series; adjacent-organ invasion common	Poor
Therapy	Multimodal if fit; VEGF-targeted therapy modest; checkpoint inhibition encouraging	SWI/SNF chromatin-remodelling defects implicated — possible future target

- **Presence or absence of sarcomatoid or rhabdoid differentiation must be reported in every RCC.**

Unclassified Renal Cell Carcinoma

- **1–5% of presumed RCC** with indeterminate features after careful analysis; many poorly differentiated, **highly aggressive, particularly poor prognosis**. The **"catch-all" category is shrinking** as molecular classification expands.

KEY POINTS: PATHOLOGY OF RENAL CELL CARCINOMA

- **CCRCC is the commonest subtype (70–80%):** proximal tubule origin, hypervascularity and necrosis; **more aggressive than other common subtypes but more responsive to immunotherapy and targeted therapy**.
- **PRCC (10–15%):** more indolent, **hypovascular, MET alterations, multifocality**.
- **ChRCC (3–5%):** generally indolent; shares histopathological features with **benign oncocyoma**.

- RCC is biologically diverse; **some subtypes need molecular testing** for diagnosis.

Concept — *Histology is biology: each subtype maps to a nephron segment, a signature genetic lesion and a characteristic behaviour — and the 2022 WHO shift from morphology to **molecular definition** exists because look-alike tumours (the old type 2 PRCC, the old collecting duct carcinoma) turned out to be several diseases with very different prognoses.*

Clinical Presentation

- **Retroperitoneal location keeps many masses silent until locally advanced. >60% of RCCs are now detected incidentally**; these are more often confined with better survival, but treating many benign or indolent tumours has had **minimal effect on RCC deaths** in the United States.

TABLE 137.10 CLINICAL PRESENTATION OF RENAL CELL CARCINOMA

Category	Features
Incidental	Most common
Localised or locally advanced disease	Haematuria; flank pain; abdominal mass; perinephric haematoma; IVC obstruction: bilateral lower-limb oedema; non-reducing or right-sided varicocele
Systemic disease	Persistent cough; bone pain; cervical lymphadenopathy; constitutional symptoms (weight loss, fever, malaise); paraneoplastic syndromes

- **Flank pain** is usually from **haemorrhage and clot obstruction**.
- **Classic triad — flank pain, gross haematuria, palpable mass — is now rare** and almost always means advanced disease ("too late triad").
- **Other markers of advanced disease**: weight loss, fever, night sweats; **cervical lymphadenopathy; non-reducing varicocele; bilateral leg oedema** (venous involvement).
- **Spontaneous perirenal haemorrhage**: >50% with **unexplained perirenal haematoma** harbour an occult tumour, usually AML or RCC; repeat CT a few months later usually gives the diagnosis.

KEY POINTS: CLINICAL PRESENTATION OF RENAL CELL CARCINOMA

- The **classic triad** (flank pain, gross haematuria, palpable mass) is **rare and often reflects advanced disease**.
- **Most RCCs are detected incidentally**; early detection associates with small tumours of limited aggressive potential.
- **Spontaneous renal haemorrhage of unclear cause is often RCC or AML**.
- **Paraneoplastic syndromes** are commoner with metastatic disease; many resolve after removal of the tumour. **Hypercalcaemia is unique in being manageable medically** (hydration, diuresis and other approaches).

- **Paraneoplastic syndromes occur in 10–20%** — RCC has more, and more diverse, syndromes than almost any other cancer; hence the old name "**internist's tumour**", now better called the "**radiologist's tumour**". Commoner in advanced disease, **almost absent with small incidental masses**.
- **Mechanism**: the kidney normally makes **1,25-dihydroxycholecalciferol, renin, erythropoietin and prostaglandins**; RCC may overproduce these and also **parathyroid hormone-related peptide, lupus-type anticoagulant, human chorionic gonadotrophin, insulin, cytokines** and inflammatory mediators.

COMPARISON — PARANEOPLASTIC SYNDROMES OF RCC

Syndrome	Frequency	Mechanism	Management / significance
Raised erythrocyte sedimentation rate	Commonest — >50% of paraneoplastic findings	Inflammatory	Marker of advanced disease
Hypercalcaemia	Up to 13%	Paraneoplastic (PTH-related peptide) or osteolytic bone metastases	Hydration → furosemide diuresis; bisphosphonates standard if renal function adequate (zoledronic acid 4 mg IV every 4 weeks; withhold in renal insufficiency); denosumab if refractory; corticosteroids, calcitonin; definitive — nephrectomy ± metastasectomy
Hypertension	Common	Tumour renin; renal artery compression or encasement; intratumoural arteriovenous fistula; less often polycythaemia, hypercalcaemia, obstruction, raised intracranial pressure	—
Polycythaemia	Common	Erythropoietin from tumour or hypoxic adjacent parenchyma	—
Stauffer syndrome (non-metastatic hepatic dysfunction)	3–20%	Interleukin-6 and other cytokines	Raised alkaline phosphatase in almost all; prothrombin time or albumin abnormal in two-thirds; bilirubin/transaminases in 20–30%; thrombocytopenia, neutropenia, fever, weight loss; exclude liver metastases; biopsy shows non-specific hepatitis; normalises after nephrectomy in 60–70%; persistence or recurrence = viable tumour, poor prognosis
Others	Rare	—	Cushing syndrome, hyperglycaemia, galactorrhoea, neuromyopathy, clotting disorders, cerebellar ataxia

- **Hypercalcaemia symptoms** are non-specific: nausea, anorexia, fatigue, **reduced deep tendon reflexes**. Hypercalcaemia from **extensive osteolytic metastases is harder to palliate** (not surgically amenable) but may respond to bisphosphonates and **focused radiotherapy**.
- **General rule:** paraneoplastic syndromes need **surgical excision or systemic therapy; medical therapy helps only in hypercalcaemia**.

Concept — Presentation has shifted from the **endocrine kidney** (triad and paraneoplastic phenomena, reflecting bulky advanced disease) to the **incidental mass**; symptoms, when present, are therefore themselves a marker of advanced stage.

Screening and Clinical Associations

- **Population screening is not feasible:** low incidence of lethal kidney cancer and no cheap, effective test — although localised RCC is highly curable and metastatic RCC is not.
- **Limiting factors:** incidence only about **16 per 100,000 per year**; about **80% of newly diagnosed small renal masses are benign or indolent** versus about **20% potentially aggressive**. A test would need

near-100% specificity, and yield would still be too low to be cost-effective. **Even high-risk groups** (male, older, heavy smokers) carry only **twofold to threefold** relative risk.

- **Dipstick haematuria** yields very little RCC — small RCCs are **parenchymal**, not urothelial, and often cause no haematuria. **Ultrasound or CT screening** finds RCC at **23–300 per 100,000** with more organ-confined tumours, but is **unlikely to be cost-effective**.
- **Candidate biomarkers** (none validated): **urinary aquaporin-1, perilipin-2**, microsatellite alterations, **VHL** mutation or hypermethylation, **VEGF, CA-IX**, serum **kidney injury molecule-1 (KIM-1)**, circulating tumour cells and **circulating tumour DNA**.
- **Screening must target defined populations: ESRD, tuberous sclerosis, familial RCC.**

TABLE 137.11 SCREENING FOR RENAL CELL CARCINOMA: TARGET POPULATIONS

Population	Recommendation
ESRD	Only if long life expectancy and few comorbidities ; periodic ultrasound or CT from the third year on dialysis
Known VHL disease	Abdominal MRI or ultrasound every 2 years from age 15–20 ; periodic screening for non-renal manifestations
Relatives of VHL patients	Genetic testing ; positive → VHL protocol; negative → less stringent follow-up
Relatives with other familial RCC	Periodic ultrasound or CT; strongly consider genetic testing
Tuberous sclerosis	Periodic ultrasound or CT
Autosomal dominant polycystic kidney disease (ADPKD)	Routine screening not justified
General population	Routine screening not justified

Screening in Cystic Kidney Disease

- **80% of ESRD patients develop acquired renal cystic disease; 1–2% of these develop RCC.** Relative risk of RCC in ESRD **fivefold to twentyfold**; **15% have metastases at presentation**.
- **Screening is recommended**, but limited by short life expectancy, **adenomas in 20–40%**, distorted anatomy and cost — **compromise: screen younger, fitter patients from the third dialysis year**.
- **Renal transplant recipients** remain at risk in **native kidneys** (RCC in 1.4–2.3% within 3 years) → **continue periodic imaging after transplantation**.
- **ADPKD**: imaging very difficult and **no significantly increased RCC risk** → **screening not recommended**.

Screening in RCC Familial Syndromes

- **Tuberous sclerosis**: adenoma sebaceum, epilepsy, intellectual disability, cysts and AMLs; RCC often early-onset and multifocal; the **Tsc2-mutant Eker rat** develops RCC frequently → **periodic renal imaging**, which also follows AML.
- **Suspect VHL** with **early-onset or multifocal RCC**, or RCC plus: **visual or neurological history; pancreatic cysts, epididymal lesions or inner-ear tumours; family history of blindness, CNS tumours or renal cancer**. Patients and relatives should consider **genetic evaluation**.

VHL ALLIANCE SURVEILLANCE PROTOCOL

Test	Start
Physical and ophthalmological examination, annually	Infancy
Plasma or urinary metanephrines, annually	Age 5
CNS MRI and audiogram, every 2 years	Age 11

Test	Start
Abdominal MRI, every other year	Age 15
MRI of internal auditory canal	Age 15

- Less intensive protocols exist, but **all relevant organ systems must be covered**.
- **Genetic counselling for all patients with RCC at ≤46 years**, given the higher likelihood of germline mutation. Carriers of *c-MET* or other relevant mutations, and those with suggestive histories, need **periodic ultrasound or CT**.

Concept — *Screening follows prevalence: in the general population the yield is too low and most detected masses are indolent, so screening is justified only where prior probability is high* — **ESRD, tuberous sclerosis and the familial syndromes** — with ADPKD the classic exclusion.

STAGING

- **Staging estimates prognosis, guides treatment and provides a common language.**
- **Robson classification** was used until the 1990s; replaced by the **tumour, node, metastasis (TNM) system** of the Union for International Cancer Control, endorsed by the **American Joint Committee on Cancer (AJCC)**. Current: **AJCC eighth edition (2016)**.

TABLE 137.12 INTERNATIONAL TNM STAGING SYSTEM FOR RENAL CELL CARCINOMA

Category	Definition
T1a	≤4.0 cm, confined to kidney
T1b	>4.0 to ≤7.0 cm, confined
T2a	>7.0 to ≤10.0 cm, confined
T2b	>10.0 cm, confined
T3a	Into renal vein or segmental branches, or pelvicalyceal system, or perirenal and/or renal sinus fat, not beyond Gerota's fascia
T3b	Grossly into vena cava below the diaphragm
T3c	Into vena cava above the diaphragm, or invades the caval wall
T4	Beyond Gerota's fascia, including contiguous ipsilateral adrenal
N1	Regional lymph node metastasis
M1	Distant metastasis
Stage I / II	T1 N0 M0 / T2 N0 M0
Stage III	T1–2 N1 M0, or T3 any N M0
Stage IV	T4 any N M0, or any T any N M1

Modified from AJCC Cancer Staging Manual, 8th edition, 2016.

TABLE 137.13 TNM STAGE ACROSS EDITIONS AND 5-YEAR CANCER-SPECIFIC SURVIVAL

Finding	Robson	TNM 6th (2002)	TNM 7th (2009)	TNM 8th (2016)	5-year survival (%)
Organ-confined (overall)	I	T1–2	T1–2	T1–2	70–90
≤4.0 cm	I	T1a	T1a	T1a	95–100
>4.0–7.0 cm	I	T1b	T1b	T1b	85–95
>7.0–10.0 cm	I	T2	T2a	T2a	65–85

Finding	Robson	TNM 6th (2002)	TNM 7th (2009)	TNM 8th (2016)	5-year survival (%)
>10.0 cm	I	T2	T2b	T2b	55–75
Pelvicalyceal invasion	I	T1–2	T1–2	T3a	55–75
Perinephric or sinus fat	II	T3a	T3a	T3a	55–75
Renal vein or branches	IIIA	T3b	T3a	T3a	45–65
IVC below diaphragm	IIIA	T3c	T3b	T3b	40–55
IVC above diaphragm or wall invasion	IIIA	T3c	T3c	T3c	20–45
Direct adrenal involvement	II	T3a	T4	T4	0–35
Beyond Gerota's fascia	IVA	T4	T4	T4	0–30
Lymph node involvement	IIIB	N1–2	N1	N1	0–30
Systemic metastasis	IVB	M1	M1	M1	0–30

Data from Amin et al., AJCC 8th edition, 2017, and other sources as cited in the source table.

- **What the table tracks** — the edition changes:
- **6th (2002): T1 split at 4 cm** (excellent outcomes ≤ 4 cm with PN or RN).
- **7th (2009): T2 split at 10 cm; ipsilateral contiguous adrenal $\rightarrow$ T4; non-contiguous adrenal (either side) $\rightarrow$ M1; isolated renal vein thrombus downgraded T3b $\rightarrow$ T3a** (favourable prognosis); N1/N2 collapsed to N1 (subdivision lacked prognostic value).
- **8th (2016): pelvicalyceal invasion $\rightarrow$ pT3a** (independent prognostic significance).
- **Trap:** compare studies across eras with care — the same tumour may carry different stages.
- **TNM limitations:** stage is set by the **single worst feature**, losing information (e.g., adrenal involvement plus supradiaphragmatic thrombus); **continuous variables made discrete** lose accuracy. Record all adverse features parenthetically; **integrated nomograms often outperform staging**.
- **Clinical staging begins with history, examination, imaging and selected laboratory tests.**
- **Suggest advanced disease:** weight loss >10%, cachexia, poor performance status; palpable mass or lymphadenopathy; **non-reducing varicocele or leg oedema** (venous involvement); anaemia, hypercalcaemia, abnormal liver function or sedimentation rate, **raised alkaline phosphatase or lactate dehydrogenase**.
- **Radiographic staging:** **high-quality abdominal CT plus chest radiograph** in most; **MRI** for locally advanced disease, poor renal function, venous involvement or contrast allergy; **chest CT** as indicated.
- **Perinephric fat: stranding is non-specific;** a distinct enhancing nodule in perinephric space is more definitive but uncommon; overall **CT/MRI accuracy for perinephric invasion is low** (microscopic spread).
- **Adrenal:** CT plus intraoperative inspection. **Enlarged or indistinct adrenal, extensive replacement with no plane, or palpably abnormal gland $\rightarrow$ treat as involved. Normal on imaging and inspection $\rightarrow$ no routine adrenalectomy.**
- **Lymph nodes:** nodes ≥ 2 cm on CT almost always malignant — confirm surgically or by biopsy if not a surgical candidate. **Smaller nodes are often inflammatory and should not preclude surgery.** MRI distinguishes vessels (lumbar veins) from nodes.
- **Invasion of adjacent organs:** **MRI is the premier study; loss of fat plane (e.g., with liver) on CT is usually not true invasion**, but be prepared to resect bowel, liver, spleen or pancreas with large infiltrative masses.
- **Venous thrombus:** **CT sensitivity 78% for renal vein and 96% for IVC thrombus.** Signs: venous enlargement, **abrupt calibre change, filling defect, collaterals. False negatives mostly right-sided** (short

vein, mass effect). **MRI is the premier study** for IVC thrombus, though multiplanar CT is likely equivalent; **intraoperative transoesophageal echocardiography** defines the cephalad extent and monitors the thrombus.

- **Metastatic evaluation minimum: chest radiograph and abdominopelvic CT or MRI.**
- **Bone scan only with raised alkaline phosphatase, bone pain or poor performance status.**
- **Chest CT** with pulmonary symptoms, abnormal chest radiograph or high-risk disease.
- **PET: guidelines recommend against** — good specificity, suboptimal sensitivity; may falsely raise concern.
- **Sestamibi SPECT/CT** under evaluation; **biopsy** of primary or metastatic sites selectively.

KEY POINTS: STAGING AND PROGNOSIS OF RENAL CELL CARCINOMA

- **TNM is the preferred staging system.**
- **T stage** reflects size and extension into renal or extrarenal structures.
- **Pathological stage is the most important prognostic factor.**
- **T4, N1 and M1 confer a poor outcome.**
- Other key pathological features: **nuclear grade, tumour size, histological subtype.**
- Clinical signs, tumour factors and laboratory findings also predict cancer-specific survival.
- **Integrated systems such as nomograms often outperform stage alone.**

Prognosis

TABLE 137.14 ADVERSE PROGNOSTIC FACTORS FOR RENAL CELL CARCINOMA

Clinical	Anatomical	Histological	Genomic
Poor performance status; systemic symptoms; anaemia ; hypercalcaemia ; raised lactate dehydrogenase, sedimentation rate , C-reactive protein, alkaline phosphatase; thrombocytosis	Larger size; venous involvement ; extension into contiguous organs including adrenal ; nodal metastases ; distant metastases and burden	High nuclear grade ; certain subtypes; sarcomatoid features ; histological necrosis ; vascular invasion; perinephric or sinus fat invasion ; collecting system invasion ; positive margin	CDKN2A alterations; DNA hypermethylation; Th2 immune signature

Data from Lane and Kattan, 2008; Sun et al., 2017; Ricketts et al., TCGA, 2018.

- **Tumour factors (stage, size, grade, subtype) have the greatest individual predictive ability**, but **integrative models are most powerful**. **Patient factors** (age, CKD, frailty, comorbidity) drive **overall survival** and must inform planning for localised RCC.
- **Adverse clinical findings in presumed localised RCC**: symptomatic presentation, paraneoplastic disease, **weight loss >10%**, poor performance status; **hypercalcaemia, anaemia and raised sedimentation rate** independently predict cancer-specific death even in localised CCRCC.
- **Pathological stage is the single most important factor.**
- **Organ-confined: 5-year survival 70–90%; perinephric fat invasion reduces survival by 10–15%.**
- **Renal sinus fat** is staged with perinephric fat as T3a but **may carry higher risk** (access to the sinus veins). **Collecting system invasion** worsens prognosis → now T3a.
- **Contiguous adrenal (T4) or non-contiguous adrenal metastasis → most eventually die of systemic disease.**
- **Venous involvement is salvageable: 5-year survival 45–69%** with tumour thrombus **if otherwise confined to the kidney**; much worse with nodes or metastases; intermediate with perinephric fat invasion.

- **Thrombus level** — prognostic value debated (selection bias); higher levels associate with more advanced disease, especially **at or above the hepatic veins**. **Venous wall invasion is as important as level → pT3c regardless of level.**
- **Major prognostic drop: T4, nodal or systemic metastasis.**
- **Node-positive: 5-year survival 5–30%, 10-year 0–5%.**
- **Systemic metastases (traditional):** 1-year survival <50%, 5-year 5–30%, 10-year 0–5%; **now median survival approaching 3–5 years** with VEGF-targeted therapy and ICIs.
- **Synchronous metastases fare worse than metachronous;** for metachronous disease, the **metastasis-free interval** reflects tempo.
- **Metastatic prognostic factors:** performance status, number and sites (**bone, brain, liver worse**), anaemia, hypercalcaemia, alkaline phosphatase, lactate dehydrogenase, thrombocytosis, sarcomatoid histology → **low, intermediate and poor risk groups** (Chapter 144).
- **Tumour size is independent**, partly through correlation with stage; larger tumours are more often clear cell and high grade. **10-year cancer-specific survival: pT1a 90–95%, pT1b 80–85%, pT2 about 75%; pT1a >95% at 5 years whether PN or RN.**
- **Nuclear grade is independent** (at least for CCRCC and PRCC). **Fuhrman grading has been replaced by ISUP grading based on nucleolar morphology.**
- **Subtype: sarcomatoid or rhabdoid, collecting duct, medullary or unclassified → poor;** CCRCC on average worse than papillary or chromophobe; **predictably indolent:** multilocular cystic clear cell neoplasm, mucinous tubular and spindle cell carcinoma, clear cell papillary tumour, some molecularly defined tumours.
- **Molecular factors:** the three pan-subtype markers (**CDKN2A, hypermethylation, Th2 signature**); Ki-67 proliferation index in CCRCC; VEGF family, adhesion molecules, **survivin — mostly still investigational.**
- **Nomograms:** the **Kattan nomogram was among the first;** older nomograms **underperform in contemporary cohorts** (stage migration, new therapies) — **must be continually revalidated.**
- **TNM versus prognostic algorithms:** TNM is an **anatomical universal language;** guidelines add **risk groups (grade plus stage)** to set follow-up and adjuvant decisions. **Multivariable tools outperform expert opinion and staging;** artificial intelligence integrating radiomics, pathomics and genomics may approach expert accuracy.

Concept — *Prognosis is governed by **anatomical extent first** — organ-confined, venous, locally invasive, nodal, metastatic form a steep gradient — with grade, subtype and size refining risk within each stage; venous thrombus is the notable exception, a high stage that remains curable when nothing else is wrong.*

Treatment of Localised Renal Cell Carcinoma

- Localised masses are common owing to cross-sectional imaging. **Clinical T1 masses were once all presumed malignant and treated by RN;** now **heterogeneous biology is recognised**, and options are RN, PN, thermal ablation (TA) and AS. Elective PN, once controversial, is standard of care.

American Urological Association Guidelines for Renal Mass and Localised Renal Cancer (2021)

- **Index patients abandoned** — too many grey zones in general health, oncological potential and function.
- **Greater focus on function:** creatinine-based eGFR and proteinuria assessment; assign CKD stage. Proteinuria and reduced baseline GFR independently predict functional decline and survival.
- **Nephrology referral:** GFR <45, confirmed proteinuria, diabetics with CKD, or expected post-intervention GFR <30 mL/min/1.73 m².

- **2021 additions:** adjuvant therapy addressed for the first time; genetic counselling indications expanded; follow-up recommendations merged in; objective AS criteria.
- **First guideline statement:** high-quality multiphase (pre- and post-contrast) cross-sectional imaging, now feasible in almost all — **contrast MRI even in severe CKD or ESRD**, as NSF with second- and third-generation gadolinium is almost non-existent.
- Largely concordant with the **American Society of Clinical Oncology, European Association of Urology, National Comprehensive Cancer Network and American College of Radiology** guidelines.

FIGURE 137.14 — AUA ALGORITHM: EVALUATION, COUNSELLING AND MANAGEMENT

Domain	Recommendations
Evaluation	Multiphase cross-sectional imaging ; comprehensive metabolic panel, blood count, urinalysis; if malignancy suspected, chest imaging and review of abdominal imaging; assign CKD stage (GFR and proteinuria)
Renal mass biopsy	Counsel on rationale, predictive values, risks, non-diagnostic rate. Consider when haematological, metastatic, inflammatory or infectious disease suspected. Utility-based: obtain when it may change management. Not required for young/healthy patients unwilling to accept its uncertainty, or older/frail patients managed conservatively regardless. Multiple cores preferred over fine-needle aspiration
Counselling	Urologist leads , multidisciplinary as needed; tumour biology and patient-specific risk (low risk of many cT1a masses); morbidities of each pathway; age, comorbidity, frailty, life expectancy; renal functional recovery, CKD, dialysis, overall survival; nephrology referral as above
Genetic counselling	All with renal malignancy aged ≤46; multifocal or bilateral masses; personal or family history suggesting a syndrome; first- or second-degree relative with renal malignancy or known syndrome; pathology suggestive of a syndrome
PN and nephron-sparing	Prioritise PN for cT1a when intervention indicated. Prioritise nephron-sparing for solitary kidney (anatomical or functional), bilateral tumours, familial RCC, pre-existing CKD or proteinuria. Consider for young patients, multifocal masses, comorbidities likely to affect future renal function
RN	Consider when increased oncological potential (size, RMB, imaging). RN preferred only if all three: (1) high complexity, PN challenging even for experts ; (2) no pre-existing CKD or proteinuria ; (3) normal contralateral kidney and new baseline GFR likely >45 after RN. Otherwise PN, unless overriding safety or oncological concerns
Thermal ablation	Alternative for cT1a solid masses <3 cm; percutaneous preferred; radiofrequency ablation (RFA) and cryoablation both options ; RMB before (preferred) or at TA; counsel on higher persistence/recurrence , treatable by repeat TA
Principles of PN	Maximise nephron mass; avoid prolonged warm ischaemia. Negative margins a priority ; extent of normal parenchyma at surgeon's discretion; consider enucleation in familial RCC, multifocal disease or severe CKD
Surgical principles	Lymph node dissection of all clinically positive nodes for staging; adrenalectomy if imaging or operative findings suggest metastasis or invasion ; minimally invasive approach when it does not compromise outcomes
Other	Pathological evaluation of adjacent parenchyma for nephrological disease; medical oncology referral for metastasis or incomplete resection; counsel high-risk or locally advanced resected patients on adjuvant therapy and trials

Scope: clinically localised masses suspicious for RCC in adults, including solid enhancing tumours and Bosniak 3 and 4 cysts. From Campbell et al., AUA Guideline Part I, J Urol 2021.

Risk Stratification and Renal Mass Biopsy

- **About 20% of solid, enhancing cT1 masses are benign** — usually oncocytoma or atypical AML. **Young to middle-aged women** have more benign pathology (**up to 40%**); **size is the stronger determinant** (about 30% of masses <2 cm benign versus about 10% of cT1b).

- **Size tracks aggressiveness:** in tumours <4 cm, **fat invasion, venous or nodal involvement and high grade are uncommon.**
- **Male sex and size are the strongest predictors of malignancy;** age, BMI and incidental presentation are inconclusive. **Clinical/radiographic algorithms: concordance <0.65 — little better than a coin toss.**
- **RMB substantially improves risk stratification** and should be considered **whenever it may influence management;** it can **spare intervention for benign tumours.**

KEY POINTS: RENAL MASS BIOPSY

- **Safe:** haematoma about 5%, clinically significant pain about 1%, gross haematuria about 1%, pneumothorax <1%, **transfusion <0.5%.**
 - **Tumour seeding almost unreported** in contemporary series.
 - **A positive biopsy is reliable: specificity 94%, positive predictive value 98.8%.**
 - **Non-diagnostic rate about 14%,** substantially reduced by **repeat biopsy.**
 - **Subtype accuracy ≥90%; grade accuracy variable (60–80%).**
 - **A non-malignant result may not mean the lesion is benign.**
- **Downsides:** non-diagnostic in about 14%, occasional false negatives, **spillage risk with cystic tumours.** **Office-based ultrasound-guided RMB by urologists** is increasing.
- **Molecular profiling of RMB** is a research priority, analogous to early prostate cancer. **Molecular imaging — anti-CA-IX antibody PET** for aggressive subtypes (CCRCC, former type 2 PRCC); **sestamibi SPECT/CT** for oncocytoma and hybrid tumours.

FIGURE 137.15 — DISTRIBUTION OF THE CLINICAL T1 RENAL MASS

Biology	Share	Matched management
Benign	About 20%	Active surveillance
Indolent RCC	About 60%	Active surveillance or thermal ablation
Potentially aggressive RCC	About 20%	Surgical excision

- The proportions **vary with size, sex and other factors;** **RMB with molecular profiling** is the AUA panel's stated research priority to allocate patients rationally.

Renal Function After Partial or Radical Nephrectomy: Survival Implications

- **Surgery remains the mainstay of cure. Robson (1969) established RN as the gold standard,** supplanting simple nephrectomy. RN is still preferred for **very large or invasive tumours,** but has **fallen out of favour for small masses because of CKD.**
- **RN predisposes to CKD,** linked to cardiovascular events and mortality. In a landmark cohort of apparently ideal elective-PN candidates, **about a quarter already had CKD (GFR <60) — RCC patients are not comparable to screened living kidney donors — and CKD was far commoner after RN than PN (65% versus 20%).**
- **CKD severity and death:** population data show **relative death rates rising steeply as eGFR falls** (about 1.2 at 45–60, 1.8 at 30–45, 3.2 at 15–30 and 5.9 at <15 mL/min/1.73 m²), after adjustment for confounders.
- **Retrospective meta-analysis favoured PN:** less severe CKD, better overall survival and better cancer-specific mortality — but **PN cannot be oncologically superior to RN, so the cancer-specific advantage exposes selection bias,** which likely inflates the overall-survival benefit too. **An Agency for Healthcare Research and Quality systematic review: cancer-specific survival depends on size and stage, not strategy; overall survival on age and comorbidity.**

LANDMARK TRIAL — EORTC 30904 (REPORTED 2011)

Element	Detail
Design	European Organisation for Research and Treatment of Cancer (EORTC) randomised trial; over 500 patients, tumours ≤5 cm, unifocal, normal contralateral kidney
Arms	RN versus elective PN
Morbidity	Lower perioperative morbidity with RN
Function	Better renal function with PN
Oncological events	Uncommon and similar
Overall survival	10-year overall survival better with RN (81% vs 76%); fewer cardiovascular deaths with RN — opposite to expectation
Verdict	Substantial flaws; primary finding widely dismissed , but suggests the functional advantage of PN in a healthy patient with a normal contralateral kidney may be smaller than believed

COMPARISON — MEDICAL VERSUS SURGICAL CKD

Feature	CKD from medical causes (CKD-M)	CKD from surgery alone (CKD-S)
Cause	Hypertension, diabetes — ongoing disease	Nephron removal — a one-time insult
Trajectory	Progressive decline (about 4.7% per year in pre-existing CKD needing surgery)	Stable (about 0.7% per year)
Survival	Reduced	Nearly identical to no CKD , especially if new baseline GFR >45

- **Functional difference between PN and RN is smaller than assumed: PN preserves about 90% of global function; RN about 65–70%**, with overlap — about half of RN patients recover to within 5% of baseline by 2 years. **Compensatory hypertrophy after RN averages 20–30%**.
- **Predicting post-RN GFR** (one of the RN criteria): nuclear renography is subjective; **parenchymal volume analysis (PVA) estimates split renal function (SRF) more accurately**:
- **New baseline GFR after RN** $\approx$ **global GFR $\times$ SRF(contralateral) $\times$ 1.25**, where 1.25 is the average adult compensation.
- Figure 137.16 example: left 223 cm³; right 234 cm³ minus 54 cm³ tumour = 180 cm³ → **SRF 55% left, 45% right**. PVA software is affordable and can be built into routine reports, as for living donors.
- **CKD and recurrence**: reports linked worse CKD to reduced recurrence-free survival (impaired immune surveillance), but multivariable work shows **tumour characteristics drive recurrence; renal function is a confounder**, because large aggressive tumours both destroy parenchyma and attract RN.

KEY POINTS: PARTIAL NEPHRECTOMY VERSUS RADICAL NEPHRECTOMY

- **PN is preferred for cT1a (≤4.0 cm) masses whenever feasible; RN is gross overtreatment** for most.
- **PN strongly preferred when function matters**: pre-existing CKD or proteinuria, abnormal contralateral kidney, multifocal or familial RCC.
- **cT1b/T2 tumours** have greater oncological potential and have often replaced much parenchyma; with a normal contralateral kidney, **PN versus RN is debatable**.
- **AUA criteria for RN** are well defined; if not met, **PN unless overriding safety or oncological concerns**.
- **Randomised trials are needed** for higher-quality data.

Radical Nephrectomy

- **Prototypical RN (Robson):** early ligation of renal artery and vein → removal outside Gerota's fascia → ipsilateral adrenalectomy → extended lymph node dissection (LND).
- **Perifascial dissection is essential** — about 25% of cT1b/T2 RCCs involve perinephric fat.
- **Early arterial ligation** remains accepted, though **collateral supply in large tumours** may prevent hilar control before mobilisation.
- **Adrenalectomy is not routine** without radiographic enlargement or invasion, unless the tumour extensively involves the kidney or is locally advanced.
- **Routine extended LND is not justified:** RCC often spreads **haematogenously**, lymphatic drainage is **variable**, and no cohort with therapeutic benefit has been defined. **LND when lymphadenopathy is suspected on imaging or at surgery; selectively in locally advanced disease — primarily for staging.**
- **Open approach** (Chapter 141): **extended subcostal** preferred by the authors; **midline** alternative; **thoracoabdominal** for very large upper-pole tumours; **extraperitoneal flank** for elderly or high-risk patients (limited exposure).
- **Laparoscopic RN** is established and less morbid for tumours **without gross local invasion, with limited or no venous involvement and manageable lymphadenopathy**, reproducing the oncological tenets; selected obese, previously operated or large-tumour patients may qualify. **Its appeal has driven overuse of RN for small masses.**
- **Recurrence after RN is stage-dependent → risk-adapted follow-up.** A post hoc analysis of the ASSURE adjuvant trial showed published recurrence algorithms perform **little better than stage and grade**, so the AUA risk groups rest on those; **histology** (especially sarcomatoid or rhabdoid) also matters.

FIGURE 137.18 — AUA FOLLOW-UP AFTER INTERVENTION: RISK GROUPS

Risk group	Definition	Chest imaging
Low (LR)	pT1, grade 1–2	Chest radiograph
Intermediate (IR)	pT1 grade 3–4, or pT2 any grade	Chest radiograph
High (HR)	pT3 any grade	CT chest preferred
Very high (VHR)	pT4 or pN1, or sarcomatoid/rhabdoid dedifferentiation, or macroscopic positive margin	CT chest preferred

- **Microscopically positive margin → move up at least one risk level.**

FIGURE 137.18 — IMAGING SCHEDULE (MONTHS AFTER SURGERY)

Risk	3	6	9	12	18	24	30	36	48	60	72–84	96–120
LR				•		•			•	•	•	•
IR		•		•		•		•	•	•	•	•
HR		•		•	•	•	•	•	•	•	•	•
VHR	•	•	•	•	•	•	•	•	•	•	•	•

- **General principles of follow-up:**
- **Benign pathology** → occasional clinical and laboratory review; **no routine imaging.**
- **Malignant** → history, examination, laboratory tests, imaging for recurrence, metastasis and treatment sequelae; **serum creatinine, eGFR and urinalysis (proteinuria) at every visit;** other bloods at discretion. **Progressive renal insufficiency or proteinuria → nephrology.**

- **Bone scan only with bone pain, raised alkaline phosphatase or suspicious imaging. Acute neurological symptoms → prompt brain/spine CT or MRI. PET not routine.**
- **Metastatic findings → define extent, refer to oncology; resect or ablate isolated or oligometastatic disease** in selected patients. **New primary or local recurrence → metastatic work-up;** if isolated to kidney or retroperitoneum, **urologist involved** in surgery or ablation decisions.
- **Abdominal imaging:** pre- and post-contrast CT or MRI preferred; **after 2 years, ultrasound may alternate** in LR and IR; **after 5 years, shared decision-making — 30% of recurrences occur beyond 60 months.**
- **After TA (biopsy malignant or non-diagnostic): contrast cross-sectional imaging within 6 months,** then the **IR protocol,** prioritising cross-sectional imaging until the ablated tumour has involuted.

Partial Nephrectomy

- **First described by Czerny (1890);** little used until the 1980s. Drivers: better imaging, renovascular surgical experience, ischaemia protection, incidental tumours, **appreciation of CKD harms,** good long-term survival.
- **Definition: complete local resection of the tumour with maximal preservation of functioning parenchyma.** Sequence (Figure 137.19): **temporary occlusion of the pedicle → excision with a rim of normal parenchyma → collecting system closure and ligation of transected vessels → capsular reconstruction.**

COMPARISON — INDICATIONS FOR PARTIAL NEPHRECTOMY

Category	Situations
Imperative (absolute)	RN would leave the patient anephric or likely dialysis-dependent: bilateral RCC; solitary functioning kidney (agenesis, prior nephrectomy, irreversible contralateral damage)
Relative	Unilateral RCC with a contralateral kidney threatened by renal artery stenosis, hydronephrosis, calculi, diabetes or severe hypertension
Elective (standard of care)	cT1a with a normal contralateral kidney, when amenable

- **Bilateral synchronous RCC: bilateral PN whenever feasible, usually staged;** if one side needs RN, **do the contralateral PN first,** then RN. **Margin width is immaterial provided the margin is negative — critical for hilar tumours.**
- **Solitary kidney:** counsel on dialysis — **temporary dialysis about 3.5%; ESRD about 4.5%** in a large series. **A remnant of at least 20–30% of one kidney** is needed to avoid ESRD. **Local recurrence 3–5% for imperative indications.**
- **When PN is not feasible for an absolute indication:** RN then dialysis (later transplantation after a cancer-free interval), or **neoadjuvant TKI to downsize** (best in clear cell). **Stop TKIs for several half-lives before and after surgery (impaired healing);** complications, mainly **urine leaks,** are commoner but manageable. **Upfront surgery whenever feasible.**
- **Elective PN for cT1a: equivalent oncological outcomes to RN,** better function; **local recurrence 1–2%, cancer-free survival >90%.** Most recurrences are **undetected multifocal disease distant from the tumour bed.** This is **balanced by a 1–2% incidence of contralateral RCC,** where RN would have left a solitary kidney with cancer.
- **PN for larger (pT1b, pT2), complex or hilar tumours** is safe and effective in selected patients; pursue whenever function matters. **Following the AUA RN criteria would greatly reduce RN overuse.**
- **Work-up:** exclude local extension and metastasis; **contrast three-dimensional volume-rendered CT (MRI in severe CKD)** to map vessels and collecting system.

- **Urological complications (urine leak, bleeding) are commoner after PN than other strategies** — a highly vascular reconstructed organ — but mostly conservatively managed. **R.E.N.A.L. and other nephrometry scores** predict complexity and complications. **Laparoscopic or robotic PN ≈ open PN** in experienced hands.

FIGURE 137.20 — DETERMINANTS OF RENAL FUNCTION AFTER PN

Determinant	Modifiable?	Weight
Quality of parenchyma before surgery	No — sets the ceiling of recovery	Fixed
Quantity of vascularised parenchyma preserved	Yes	Primary determinant
Ischaemic injury	Yes	Secondary, if warm ischaemia is limited or hypothermia used; prolonged warm ischaemia → irreversible loss

- **Technique: precise excision with a small rim, careful reconstruction minimising devascularisation, facilitated by a short ischaemic interval.**
- **Hypothermia** when ischaemia is expected to be **prolonged (≥25–30 minutes)**, especially with a **solitary kidney or CKD**.
- **Segmental clamping, zero-ischaemia PN and three-dimensional printing** are promising. **Randomised off-clamp versus on-clamp trials show no functional advantage for off-clamp** (comparator was limited warm ischaemia). Susceptible cohorts and the role of acute kidney injury remain debated.
- **Hyperfiltration after PN in a solitary kidney: >50% reduction in parenchymal volume → proteinuria, focal segmental glomerulosclerosis and progressive renal failure;** proteinuria rises with follow-up time and falls with remnant size. **Mitigation: angiotensin-converting enzyme inhibitors plus low-protein diet.**

Tumour Enucleation

COMPARISON — STANDARD PARTIAL NEPHRECTOMY VERSUS TUMOUR ENUCLEATION (FIGURE 137.22)

Feature	Standard PN	Tumour enucleation (TE)
Dissection	Strategic sharp dissection through parenchyma	Blunt dissection along the pseudocapsule
Margin	2–3 mm rim of normal parenchyma	Pseudocapsule, little or no parenchyma
Advantages	Avoids the pseudocapsule — oncological orthodoxy	Less blood loss; avoids collecting system entry ; facilitates zero ischaemia ; may obviate capsular closure; maximal parenchyma
Function	Already saves about 90% of global function	Small increment — benefit mainly in severe pre-existing CKD
Concern	Sharp dissection is "blind" at infiltrated zones	Positive margins — 40–50% of localised RCC infiltrate into or beyond the pseudocapsule

- **Established role: familial RCC; for sporadic RCC controversial. Positive margins after PN increase recurrence,** yet most TE series report **low positive-margin rates and good outcomes**. Proponents argue the **inflammatory reaction around infiltrated pseudocapsule steers blunt dissection away** — "**keep your friends close, and your enemies closer**"; alternatives are differing margin assessment and **selection of homogeneous, encapsulated tumours**. A randomised trial with uniform pathology is needed.
- **TE is probably performed more often than recognised.** Record technique — **enucleation, standard resection or enucleoresection** — from the gross specimen at the **surface, intermediate region and base (SIB protocol)**.

- **Capsular invasion is less common in low-grade tumours, ChRCC and oncocytoma; high-grade tumours risk recurrence after TE, so RMB can assess suitability. Negative margins are always the top priority; the extent of normal parenchyma removed is at surgeon's discretion.**

Thermal Ablative Modalities

- **Cryoablation and RFA are established nephron-sparing alternatives for small localised tumours, usually percutaneous, with a favourable morbidity profile relative to PN and limited effect on renal function — reasonable for a small tumour in a solitary kidney, though PN remains preferred there.**
- **Oncological efficacy is strong, but excision remains the reference standard: local recurrence is higher after primary TA (Figure 137.23: meta-analysis risk ratio about 0.55 favouring PN, with no difference once secondary ablations are counted). Other concerns: RMB may miss heterogeneity; no pathological staging. Most local recurrences are salvaged by repeat ablation; cancer-specific survival for cT1a is high whatever the strategy; overall survival depends on age and health.**
- **Candidates: elderly or comorbid patients wanting proactive treatment (AS now often used instead); local recurrence after PN; hereditary multifocal RCC; informed patient preference.**
- **Size: success highest and complications lowest below 2.5–3.0 cm; local recurrence-free rates >90% in some centres. Lower success with avidly enhancing masses and clear cell histology → preliminary RMB. Location near hilar vessels, bowel, pancreas or spleen raises risk — hydrodissection (saline) or laparoscopic mobilisation may help.**
- **AUA 2021: TA is an alternative for cT1a masses <3.0 cm, even in healthy patients; RMB before TA preferred.**

COMPARISON — CRYOABLATION VERSUS RADIOFREQUENCY ABLATION

Feature	Cryoablation	Radiofrequency ablation
Experience	Longer and more extensive	Initially more variable; now comparable
Mechanism	Intracellular ice disrupts membranes and organelles → delayed microcirculatory failure and anoxia	High-frequency current → ionic agitation, friction, heat → protein denaturation and membrane melting
Lethal threshold	–20 °C or colder, reached 3–5 mm inside the leading edge of the iceball	Effects above 41 °C , rising with temperature and time; >100 °C at probe tip , dissipating rapidly
Technique requirements	Rapid freeze, slow thaw, repeat the freeze–thaw cycle; extend the iceball about 1 cm beyond the tumour	Multiple probes or tines ; thermosensors in areas of concern
Monitoring	Real-time iceball on ultrasound	No "iceball equivalent" — relies on empirical probe alignment
Local control / recurrence	Control about 80–95%; persistence or recurrence 3–10%	Persistence or recurrence 5–20% ; low for tumours <3 cm
Complications	Renal fracture, haemorrhage, adjacent organ injury, ileus, wound infection	Acute renal failure, ureteropelvic junction stricture, necrotising pancreatitis, lumbar radiculopathy
AUA verdict	Neither superior, given adequate expertise	

- **Recurrence after TA: central or nodular enhancement in the ablation bed on extended follow-up is considered diagnostic; also growth, new nodularity, failure to regress, satellite or port-site lesions → biopsy and consider retreatment. Few studies used routine post-ablation biopsy.**
- **Salvage: usually repeat ablation; salvage surgery is hampered by dense fibrosis, sometimes precluding PN or minimally invasive surgery.**

- **Developmental modalities:** high-intensity focused ultrasound, **stereotactic body radiotherapy**, microwave ablation, laser interstitial thermal therapy, **irreversible electroporation** — complete cell kill not yet established.

Active Surveillance

- **Early AS series:** small (≤ 3.5 cm) solid enhancing masses grew about **0.36 cm per year**; those excised after exceeding 3 cm were **organ-confined, mostly low grade; no metastases**.
- **Pooled series:** median growth **0.09–0.34 cm per year**; metastasis **<2% over 2–5 years**.
- **Limitations:** **selection bias** (small, homogeneous masses); **20% or more may have been benign** (no RMB); a **subgroup grows rapidly** (in one series a quarter doubled in volume within a year), and delay may forfeit the window for PN or TA.
- **Elderly or high-risk patients** can be safely observed with imaging every **6–12 months**; **competing mortality and intervention risk often exceed RCC risk**, and treating small masses in the elderly may confer **no survival benefit**.
- **Prospective data support AS even in healthy patients** with low-potential masses (≤ 2 cm, incidental).

LANDMARK REGISTRY — DELAYED INTERVENTION AND SURVEILLANCE FOR SMALL RENAL MASSES (DISSRM)

Element	Detail
Population	Incidental, solid, small renal masses; AS versus primary intervention by choice
Follow-up	Median 3.3 years , many beyond 5 years
Outcome on AS	Only one patient died of RCC or developed metastasis
Delayed intervention	About 24% (growth or preference)

- **Predominantly cystic masses** are less aggressive → consider AS. **Delayed intervention is usually salvageable with nephron-sparing surgery**. Some advocate **AS as the default initial strategy** for most small masses.

FIGURE 137.24 — AUA ACTIVE SURVEILLANCE ALGORITHM

Element	Recommendation
Eligibility	Solid mass <2 cm, or complex but predominantly cystic → AS is an option for initial management (strong consideration); small solid or Bosniak 3–4 masses → AS an option
Prioritise AS / expectant management	When intervention risk or competing mortality outweighs oncological benefit ; if asymptomatic, surveillance by shared decision-making
Equivocal risk–benefit, patient prefers AS	Repeat imaging at 3–6 months , consider RMB; again 3–6 months later; then by growth rate — intervene if substantial growth or the balance shifts
Benefit of intervention clear	Recommend intervention ; AS only if the patient accepts the risk; encourage RMB ; close surveillance and counselling
Patient factors favouring AS	Elderly ; life expectancy <5 years ; high comorbidity; excessive perioperative risk; frailty ; preference; marginal renal function
Tumour factors favouring AS	Size <3 cm ; growth <5 mm per year ; non-infiltrative; low complexity; favourable histology; predominantly cystic
Frequency	AS: about every 3–6 months , cross-sectional imaging and/or ultrasound. Expectant management: every 6–12 months , more ultrasound
Triggers for change	Size >3 cm ; stage progression ; growth >5 mm per year ; clinical change; new biopsy result; symptoms ; metastatic progression

Element	Recommendation
Adjuncts	Concurrent renal function (creatinine, proteinuria), liver function tests, chest imaging ; contrast alternatives (Doppler, diffusion-weighted MRI) when needed

- **AS versus expectant management:** AS implies **intent to intervene on progression**; expectant management is **less intensive, often palliative in intent**.

Nephron-Sparing Surgery in VHL and Other Familial RCC

- **VHL RCC:** younger, multiple bilateral tumours, generally low stage but **can metastasise and frequently cause death**. Histology: **solid clear cell tumours plus cysts lined by carcinoma or hyperplastic clear cells (incipient carcinoma)**.
- **Options:** bilateral nephrectomy with renal replacement, or **nephron-sparing PN or TA to avoid ESRD** — preferred, **even for central tumours**; **enucleation favoured** to preserve parenchyma.
- **Ipsilateral recurrence is frequent** (occult microscopic disease). In a multicentre VHL series, **cancer-specific survival was high at 5 years and fell by 10 years**, and **freedom from ipsilateral recurrence after PN fell steeply between 5 and 10 years** — **long-term surveillance is mandatory**.
- **The 3 cm rule** (National Cancer Institute): **no metastases in solid VHL tumours <3 cm** observed for about 5 years, versus about a quarter metastasising above 3 cm, rising with size. **Withhold PN until the tumour reaches 3.0 cm**. Applies also to HPRC and BHD; **HLRCC is the exception** — **treat aggressively even when <3 cm**.
- **Most VHL patients develop recurrence needing re-intervention**; repeat PN is difficult (fibrosis), so **TA** may reclaim control.
- **Belzutifan (oral HIF-2α inhibitor)** slows growth and reduces interventions: **renal response 30–50%**, most tumours shrink or stabilise; **commonest side effects anaemia and fatigue**.
- **Transplantation after removal of all renal tissue** is generally safe despite the tumour diathesis.
- **Before renal surgery in VHL, identify pheochromocytoma and CNS haemangioblastoma**. Genetic counselling and screening for other manifestations are part of comprehensive care.

Concept — *Management of the localised mass is a **three-way balance of oncological risk, renal function and competing mortality**: small, indolent tumours are oversold by RN, so the guidelines push towards PN, ablation or surveillance and reserve RN for tumours whose aggression, complexity and a healthy contralateral kidney all justify giving up nephrons.*

Treatment of Locally Advanced Renal Cell Carcinoma

Inferior Vena Cava Involvement

- **Intraluminal growth into the renal vein and IVC, as far as the right atrium or beyond**, is characteristic; many such patients have no metastases. **45–70% of patients with IVC thrombus can be cured by RN with IVC thrombectomy**.
- **Venous involvement in 4–10% of RCC**.
- **Suspect IVC thrombus with:** lower-limb oedema; isolated right-sided or non-collapsing varicocele; dilated superficial abdominal veins; proteinuria; pulmonary embolism; right atrial mass; non-functioning kidney.

COMPARISON — IVC THROMBUS LEVEL VERSUS TNM CATEGORY, AND OPERATIVE APPROACH

Level	Extent	TNM	Vascular control (Figure 137.27)
Renal vein only	Renal vein or branches	T3a	Standard RN

Level	Extent	TNM	Vascular control (Figure 137.27)
I	Adjacent to the renal vein ostium	T3b	Satinsky clamp or vascular stapler; preserve ≥50% of caval lumen; patch graft if needed
II	Up to the lower liver, below the hepatic veins	T3b	Sequential clamping: caudal IVC → contralateral renal vein → cephalad IVC; occlude lumbar veins; open ostium; extract in a bloodless field
III	Intrahepatic IVC, below the diaphragm	T3b	Mobilise the liver, milk thrombus below the hepatic veins, then as for level II; or clamp above the liver with a Pringle manoeuvre; venovenous bypass often (not if good collaterals)
IV	Above the diaphragm	T3c	Cardiopulmonary bypass with hypothermic circulatory arrest (still preferred in complex cases); increasingly avoided — if thrombus can be brought below the atrium, control without opening the heart or advanced venovenous bypass
Any level, wall invasion	Caval wall	T3c	Resect involved cava to negative margins; graft or reconstruct as needed

- **Level and TNM are distinct systems;** hence debate. Two large multicentre cohorts confirm **thrombus level independently predicts survival and recurrence**. **Microscopic caval wall invasion strongly predicts cancer-specific survival;** nodes, metastases, grade, non-clear-cell histology, anaemia and BMI also matter.
- **Level IV thrombi are curable** in a significant proportion without metastases; with modern systemic therapy **5-year overall survival can reach 40%.**
- **Imaging:** as close to surgery as possible (thrombus may progress). **MRI preferred at many centres; good CT is essentially equivalent; enhancement distinguishes tumour thrombus from bland thrombus.** Venacavography only if MRI/CT equivocal or contraindicated. **Thrombus is vascularised in 35–40%.**
- **Preoperative renal artery embolisation: not recommended routinely** — no measurable benefit and **more major complications and mortality.**
- **Coronary angiography before planned cardiopulmonary bypass with deep hypothermic arrest** (levels III–IV), to allow revascularisation at bypass. **Intraoperative echocardiography:** thrombus extent, emboli, cardiac function, residual tumour.
- **Sequence:** early arterial ligation, then mobilise the kidney until attached only by the renal vein; alternatively, deal with the thrombus **before mobilisation** to reduce embolism.
- **Complete caval occlusion** may not need reconstruction (**collaterals established**). **Distal bland thrombus:** remove if safe; if left, **ligate the IVC above it or place a filter** to prevent pulmonary embolism.
- **Hypothermic circulatory arrest drawbacks:** post-bypass **coagulopathy, stroke and myocardial infarction.**
- **Mortality 5–10%** for supradiaphragmatic thrombi in some series. **Select carefully** (metastatic patients have limited survival), but surgery also **palliates:** prevents pulmonary emboli, relieves oedema, ascites, cardiac dysfunction, pain and haematuria.
- **Robotic nephrectomy with IVC thrombectomy** is feasible for **selected level I–III;** oncological data immature. **Stereotactic ablative radiotherapy** for inoperable, palliative or neoadjuvant use: **up to 60% response,** mostly mild adverse events.

KEY POINTS: TREATMENT OF LOCALLY ADVANCED RENAL CELL CARCINOMA

- **45–70% of patients with venous tumour thrombus can be cured** by nephrectomy and thrombectomy, including atrial extension.
- **Thrombus below the main hepatic veins:** isolate the involved vasculature and remove the thrombus.

- **Thrombus above the main hepatic veins:** more extensive dissection, **venovenous bypass or cardiopulmonary bypass with circulatory arrest.**
- **cT4 with suspected invasion: complete en bloc excision is the only chance of cure.**
- **Bulky lymphadenopathy** carries a prognosis similar to metastatic disease, but resection should be considered if feasible.
- **High-quality CT or MRI close to the operation date** is essential.
- **Adjuvant systemic therapy is FDA approved;** trials encouraged; **neoadjuvant therapy** in selected patients.

Locally Invasive Renal Cell Carcinoma

- **T4 tumours are uncommon (<2% of surgical series)** and usually present with **pain** from posterior abdominal wall, nerve root or paraspinal muscle invasion.
- **Large tumours metastasise more readily than they invade: only about 40% of suspected direct invasion is confirmed pathologically** — classic scenario: **upper-pole tumour "invading" the liver with intact planes at exploration.**
- **Differential of a large invasive retroperitoneal mass: adrenocortical carcinoma, urothelial carcinoma, sarcoma, lymphoma,** as well as invasive RCC.
- **Surgery is the only curative option → en bloc resection** of bowel, spleen, distal pancreas or abdominal wall. Prepare: **bowel preparation, vaccination for splenectomy, endocrine replacement if both adrenals involved.**
- **Prognosis remains poor** even with complete resection — most die of disease within about a year, and median survival is far shorter with metastases → **neoadjuvant systemic therapy is a valid "litmus test"** for potentially unresectable RCC.
- **Debulking (incomplete excision) is rarely indicated** — **1-year survival only 10–20%.** Radiotherapy traditionally has little role (bowel toxicity), though **stereotactic techniques** may change this.

Lymph Node Dissection for Renal Cell Carcinoma

- **Nodal metastasis defines a high-risk group;** LND stages accurately, but **therapeutic benefit is controversial.**
- **Only 30–43% of radiographically enlarged nodes are malignant.** In high-risk patients, the chance of malignancy rises with node size (about **20% at 7 mm, 29% at 10 mm, 90% at 30 mm**). Node positivity is also **stage-specific.**

TABLE 137.15 RISK OF REGIONAL LYMPH NODE METASTASES BY NUMBER OF PATHOLOGICAL RISK FACTORS

Number of risk factors*	Share of patients	Node-positive, retrospective series†	Node-positive, prospective series‡
0	44%	0.4%	—
1	18%	1.0%	—
2	17%	4.4%	20%
3	13%	12%	37%
4	7.3%	13%	49%
5	0.9%	53%	50%

\ *Risk factors: (1) primary tumour ≥ 10 cm; (2) clinical T3/T4; (3) high grade (grade 3 or 4); (4) sarcomatoid features; (5) histological tumour necrosis. †Blute et al., J Urol 2004 (LND in 58% of 1652). ‡Crispen et al., Eur Urol 2011 (LND in 41% of 415 with ≥ 2 risk factors).

- **What the table is for:** the **five risk factors** select patients for LND — **≥ 2 factors $\rightarrow$ $>40\%$ nodal involvement** in the prospective cohort. **Body text versus table:** the retrospective and prospective columns differ at the same risk count because the **prospective cohort was selected (≥ 2 factors) and more thoroughly dissected** — a different denominator, not a contradiction.
- **Template problem:** **45% of positive nodes lie away from the renal hilum** (Figure 137.28), so hilar dissection alone under-stages.

LANDMARK TRIAL — EORTC 30881

Element	Detail
Question	Lymphadenectomy at RN
Result	No survival advantage
Criticism	Underpowered, low-risk population (81% grade 1–2; 72% organ-confined); only 4% node-positive

- **Large retrospective series:** no consistent therapeutic benefit in non-metastatic RCC (Mayo analysis, including intermediate- and high-risk cohorts); a European series suggested benefit with **bulky tumours and sarcomatoid features**; rare **isolated nodal disease** may be durably controlled by resection.
- **Bottom line:** LND for clinically suspicious nodes (radiographic or intraoperative), for staging; not routinely for localised, clinically node-negative RCC.

Local Recurrence After Radical Nephrectomy or Nephron-Sparing Surgery

COMPARISON — LOCAL RECURRENCE BY PRIMARY TREATMENT

Feature	After RN	After PN	After TA
Sites	Renal fossa, ipsilateral adrenal, renal vein stump or IVC , ipsilateral retroperitoneal nodes	Remnant kidney, usually distant from the tumour bed	Ablation zone (incomplete eradication)
Incidence	2–4%; rare if organ-confined	1.4–10%	Highest after a single treatment: cryoablation 3–10%, RFA 5–20%
Risk factors	Locally advanced or node-positive disease; adverse histology	Advanced T stage, high grade; positive margins in high-grade or high-stage disease	Size, location, complexity
Isolated?	Only 20–40% isolated — full metastatic work-up	—	—
Management	Resection of isolated recurrence $\rightarrow$ long-term cancer-free status in 30–50%; often en bloc with adjacent organs	Repeat PN, completion nephrectomy, TA or AS	Repeat TA, AS or salvage surgery (hampered by fibrosis)

- **After RN:** resection is formidable (lost tissue planes, frequent organ invasion). **Favourable features: small, unifocal recurrence; multifocal recurrence associates with distant metastasis.** Recurrence reflects **tumour biology, not surgical approach** (minimally invasive or PN). **Intraoperative radiotherapy gives no benefit; radiotherapy palliates** inoperable symptomatic recurrence.
- **Positive margins after PN:** once thought inconsequential; **now linked to worse recurrence-free and metastasis-free survival in high-grade or high-stage disease.** Usually managed by close surveillance; completion nephrectomy only if grossly involved margins with a healthy contralateral kidney.

- **After TA:** despite more local recurrence, **no difference in metastatic recurrence or cancer-specific survival** with appropriate selection.

Neoadjuvant and Adjuvant Therapy for Renal Cell Carcinoma

- **Recurrence after apparently curative surgery reflects occult micrometastases;** overall, **distant metastasis in 20–35% and local recurrence in 2–5%** → rationale for adjuvant therapy in high-risk patients.
- **Historically negative adjuvant trials:** IL-2, interferon- α , medroxyprogesterone acetate, radiotherapy, autologous tumour vaccines, and girentuximab (anti-G250 / CA-IX chimeric antibody).

LANDMARK TRIALS — ADJUVANT THERAPY

Trial	Population	Agent vs comparator	Endpoint result	Verdict
S-TRAC (phase 3)	615 patients, high-risk clear cell RCC: pT2 high grade, pT3–4 or N1	Sunitinib vs placebo	Disease-free survival longer (6.8 vs 5.6 years); no overall survival difference	Positive for DFS; other adjuvant TKI trials negative
KEYNOTE-564	RCC at high risk of recurrence (about 500 per arm)	Pembrolizumab vs placebo	Disease-free survival hazard ratio 0.63; 2-year DFS 75% vs 67%; survival 91% vs 86% as reported	Positive → FDA approval 2021; greatest benefit when metastatic disease resected to R0; only positive ICI adjuvant trial so far

- **Neoadjuvant therapy** (VEGF-targeted agents or checkpoint inhibitors) may **facilitate surgery or eradicate micrometastases**:
- **TKIs** (pazopanib, sunitinib, sorafenib, axitinib): **primary volume reduction 10–30%**; may enable resection in 40–45%, including PN candidates and thrombi; **about 20% have a serious adverse event, mostly wound healing**.
- **Neoadjuvant nivolumab** (three or four cycles): safe, surgery unaffected, **robust immunological responses but minimal tumour shrinkage**; several neoadjuvant trials ongoing.

KEY POINTS: TREATMENT OF LOCAL RECURRENCE AND ADJUVANT THERAPY

- **Isolated local recurrence after RN: 2–4%**; full metastatic work-up; if negative, **excision** (often en bloc).
- **Local recurrence after PN is usually distant from the tumour bed**; options: repeat PN, completion nephrectomy, TA or AS.
- **Local recurrence after TA usually reflects incomplete eradication**; options: repeat TA, AS or salvage surgery.
- After a long history of failure, **some recent trials show benefit from adjuvant systemic therapy in high-risk RCC**.
- **Adjuvant and neoadjuvant trials should be supported.**

Concept — *In locally advanced RCC, surgery remains the only curative act — thrombectomy at any level, en bloc resection for T4, resection of isolated recurrence — while lymphadenectomy stages rather than cures, and systemic therapy is now moving from salvage into the adjuvant and neoadjuvant space.*

OTHER MALIGNANT RENAL TUMOURS

Sarcomas of the Kidney

- **1–2% of adult renal malignancies**, peak in the **fifth decade**; **less common but more lethal than sarcoma at any other genitourinary site**.
- **Sarcoma versus sarcomatoid RCC** is often hard to distinguish; **any feature of an RCC subtype excludes primary renal sarcoma**.
- **Presentation** mirrors large RCC (mass, pain, haematuria). **Features favouring sarcoma: origin from the capsule or sinus; large size without lymphadenopathy; fat or bone** (liposarcoma, osteosarcoma); **hypovascularity** — except **haemangiopericytoma** (highly vascular). **Suspect with any very large or rapidly growing mass**.
- **Biology**: mesenchymal, lacking natural barriers; **pseudocapsule infiltrated by tumour**, extending unseen into surrounding tissue → **frequent local recurrence even after wide excision**. **High-grade: lung metastases, death within months; low-grade: indolent with repeated local recurrence**.
- **Key prognostic factors: margin status and grade**. **The first resection is the best chance of cure** → often RN with en bloc excision of adjacent organs; MRI for planning; intraoperative margin assessment.
- **Chemotherapy**: traditionally **doxorubicin and ifosfamide**; gemcitabine and docetaxel; TKIs investigational — **responses disappointing**.
- **Radiotherapy**: chemoradiation effective adjuvantly in extremity sarcoma but **less so retroperitoneally**; **preoperative radiotherapy may reduce positive margins**; **intraoperative radiotherapy** shows some benefit. **Multimodal approach if fit**.
- **Subtypes: leiomyosarcoma commonest (50–60%)**, then liposarcoma — **the reverse of retroperitoneal sarcoma, where liposarcoma is commonest**. Outcomes in all series are poor, with most dying within months to a few years.
- **Leiomyosarcoma**: from **smooth muscle of the capsule** or perinephric structures; **female predominance, fourth to sixth decades**; **displaces rather than invades** parenchyma; histology shows **spindle cells with blunt-ended nuclei and eosinophilic cytoplasm**; rapid growth, frequent metastasis and recurrence; most die within 2 years of RN.
- **Liposarcoma**: fat distinguishes it from RCC, but **confused with AML or lipoma**; grows very large; **adjuvant radiotherapy and cisplatin-based chemotherapy** for high grade or positive margins.
- **Osteogenic sarcoma**: calcified, **rock hard**; **extensive calcification in a large hypovascular mass**; plain film may **mimic a staghorn calculus** (mass effect suggests xanthogranulomatous pyelonephritis or, rarely, osteosarcoma).
- **Others**: rhabdomyosarcoma, fibrosarcoma, malignant fibrous histiocytoma; **malignant haemangiopericytoma** — **highly vascular**; **preoperative embolisation** may help.

TABLE 137.16 CHARACTERISTICS OF OTHER MALIGNANT RENAL TUMOURS

Tumour	Characteristics	Management
Sarcoma	Leiomyosarcoma commonest ; hypovascular; rapid growth; may arise from capsule	Wide local excision, confirmed negative margins
Lymphoma and leukaemia	Multiple patterns; hypovascular; massive retroperitoneal or distant lymphadenopathy, atypical sites	Biopsy; avoid extirpative surgery; chemotherapy ± radiotherapy
Metastases	Lung, breast, gastrointestinal, melanoma, haematological ; hypovascular, multifocal	Biopsy strongly considered; systemic therapy or palliation

Tumour	Characteristics	Management
Carcinoid	Hypovascular; carcinoid syndrome possible; neuroendocrine	Surgical excision
Small cell carcinoma	Neuroendocrine; hypovascular; mostly advanced	Surgery plus platinum-based chemotherapy
Primitive neuroectodermal tumour (PNET)	Neural crest ; hypovascular; poor prognosis	Multimodal
Wilms tumour	Heterogeneous solid mass; mainly children, occasionally adults	Multimodal, as paediatric protocols

- **The pattern: almost every non-RCC malignancy is hypovascular** — hypervascularity points to RCC (or haemangiopericytoma). **Lymphoma and metastasis are the two to biopsy rather than resect.**

Renal Lymphoma and Leukaemia

- **Renal involvement found at autopsy in about one-third** of those dying of lymphoma or leukaemia, but **clinically uncommon** — usually silent and late. **Urologist's role: differentiate from other tumours, obtain tissue, preserve renal function.**
- **Non-Hodgkin > Hodgkin; diffuse > nodular. Primary renal lymphoma is rare** (little native lymphoid tissue). **Haematogenous spread in about 90%**, the remainder by **direct extension from retroperitoneal nodes**. Commonest pattern: **multiple small interstitial nodules** that coalesce, and may replace the kidney → renal failure.
- **CT is the modality of choice** for diagnosis and response; **MRI** for renal insufficiency, contrast allergy or the young.

TABLE 137.17 CT FINDINGS ASSOCIATED WITH RENAL LYMPHOMA

Finding	Incidence (%)
Multiple renal masses	45
Invasion from enlarged retroperitoneal nodes	25
Solitary renal mass (difficult to separate from RCC)	15
Diffuse renal involvement	10
Predominantly perinephric	5

From Heiken *et al.*, 1983, and Pollack *et al.*, 1987.

- **Suspect lymphoma with massive retroperitoneal lymphadenopathy, splenomegaly, distant or atypical lymphadenopathy. RCC landing zones: interaortocaval for right-sided and para-aortic for left-sided tumours** — nodes centred elsewhere suggest lymphoma. **A prior lymphoma plus a renal mass → recurrence until proven otherwise.** Risk increased by **immunosuppression, acquired immunodeficiency syndrome, autoimmune disease, graft-versus-host disease, prior radiotherapy.**
- **Leukaemia** involves the kidney more in **children**, mostly **lymphocytic**, diffusely infiltrative, late.
- **Management: percutaneous biopsy or aspiration**; at exploration, **frozen section first**; **avoid extirpative surgery. Chemotherapy ± radiotherapy** — **CHOP (cyclophosphamide, doxorubicin, vincristine, prednisolone)**, with rituximab (R-CHOP) improving renal function and survival. **Nephrectomy only for severe symptoms (uncontrollable haemorrhage)**, or rare **primary renal lymphoma** (nephrectomy plus chemotherapy); some renal **mucosa-associated lymphoid tissue (MALT) lymphomas** cured by surgery alone.

- **Presentation:** usually silent; haematuria, flank pain or renal failure; **B symptoms (fever, weight loss, fatigue) are more common.** Renal failure from replacement or bilateral ureteral obstruction, but more often hypercalcaemia or urate nephropathy from tumour lysis syndrome.

Metastatic Tumours

- **Metastases are the commonest malignant renal neoplasms: 12% of patients dying of cancer have renal metastases at autopsy.** Almost all haematogenous (rich vascularity); direct invasion (pancreas, colon, adrenal) less common.
- **Sources:** lung, breast, gastrointestinal, melanoma, haematological — lung commonest among solid tumours. Usually multifocal with widespread disease, but lung, breast and colon metastases may be large and solitary, mimicking RCC.
- **CT:** isodense, only moderately enhancing (5–30 HU). Suspect with multiple lesions plus metastases or known non-renal cancer; percutaneous biopsy confirms.
- **Management:** systemic therapy or palliation; nephrectomy almost never (except haemorrhage refractory to embolisation).
- **Trap:** a solitary, strongly enhancing mass with a history of organ-confined non-renal cancer is more likely RCC, especially after a long interval — in a consecutive series, no patient without other evidence of progression had a renal metastasis.

Other Malignant Tumours of the Kidney

COMPARISON — NEUROENDOCRINE, EMBRYONAL AND ADULT WILMS TUMOURS

Feature	Carcinoid	Small cell carcinoma	PNET	Adult Wilms tumour
Frequency	<60 cases reported	About 30 primary cases	Rare in adults	3% of Wilms tumours are adult
Association	Horseshoe kidney (82-fold relative risk)	—	Ewing sarcoma family	20% at 15–20 years, 80% in third to seventh decades
Markers	Neuron-specific enolase, chromogranin; urinary/plasma serotonin	Neuron-specific enolase, chromogranin, synaptophysin	CD99, vimentin, cytokeratin, neuron-specific enolase; Homer-Wright rosettes; t(11;22)(q24;q12)	Triphasic: blastema, epithelium, stroma
Presentation	Median age 49; carcinoid syndrome in a minority (flushing, wheezing, diarrhoea); metastases at diagnosis in 46%	Mostly locally advanced or metastatic; flank pain, haematuria	Heterogeneous necrotic mass, equivocal enhancement; early local and nodal, lung, liver, bone spread	Heterogeneous hypovascular mass, usually unsuspected
Treatment	Surgical excision; nephron-sparing if suspected	Debulking plus platinum-based chemotherapy	Debulking + Ewing-type chemotherapy + renal bed radiotherapy	Multimodal, as paediatric protocols
Prognosis	Generally good, especially in horseshoe kidney	Poor	Poor; 5-year disease-free survival 45–55%	Worse than in children (advanced stage, abrupt decline in performance status)

- **Carcinoid adverse factors:** age >40, size >4 cm, high mitotic rate, purely solid, metastasis at diagnosis, extension through the capsule.

- **Small cell carcinoma** must be distinguished from **Wilms tumour, PNET, lymphoma and metastatic pulmonary small cell carcinoma**; **hypovascularity** may be a clue. Large cell neuroendocrine carcinoma also occurs rarely.
- **PNET** behaves **more aggressively in the kidney** than elsewhere; the **t(11;22) translocation** separates it from **neuroblastoma and adult Wilms tumour**.
- **Wilms tumour** is the **commonest abdominal malignancy of childhood**; **adult pathological staging is the same as paediatric**; differentiation from RCC is often impossible preoperatively.

Concept — *The non-RCC malignancies share two warning signs — **hypovascularity and an atypical pattern** (infiltrative, multifocal, massive adenopathy) — and the key decision is whether the lesion is surgical (sarcoma, carcinoid, primary RCC mimic) or medical (lymphoma, leukaemia, metastasis), which is why biopsy is favoured whenever the latter is suspected.*